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(54) **METHODS AND COMPOSITIONS FOR TREATING YELLOW FEVER**

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(65) **Prior Publication Data**

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A61K 39/00 (2006.01)
A61P 31/14 (2006.01)

(52) **U.S. Cl.**
CPC **C07K 16/1081** (2013.01); **A61P 31/14** (2018.01); **A61K 2039/505** (2013.01); **C07K 2317/41** (2013.01); **C07K 2317/76** (2013.01); **C07K 2317/92** (2013.01); **C07K 2317/94** (2013.01)

(58) **Field of Classification Search**

CPC **C07K 16/1081**; **C07K 2317/41**; **C07K 2317/76**; **C07K 2317/92**; **C07K 2317/94**; **A61P 31/14**; **A61K 2039/505**
See application file for complete search history.

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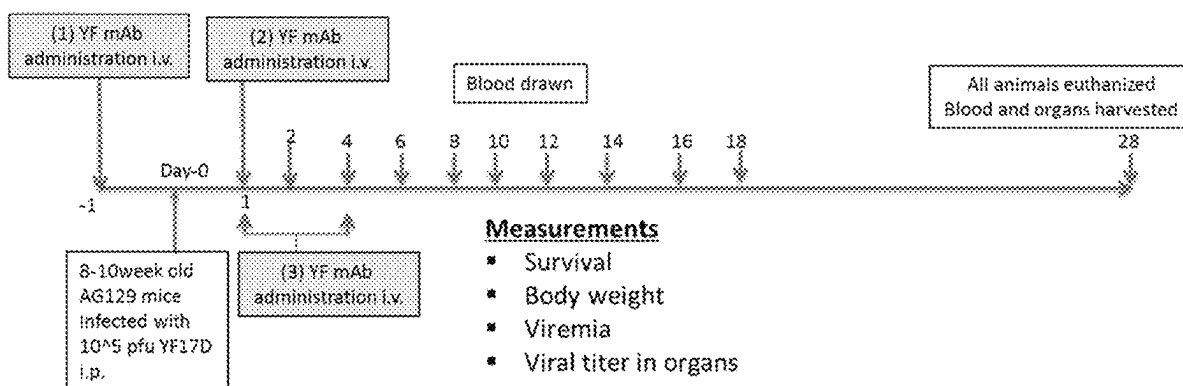
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(57) **ABSTRACT**

The present disclosure is based, at least in part, on the discovery of antibodies that specifically bind YFV and/or neutralize the virus. Compositions and methods related to such antibodies or antigen-binding portions thereof are provided.

17 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.



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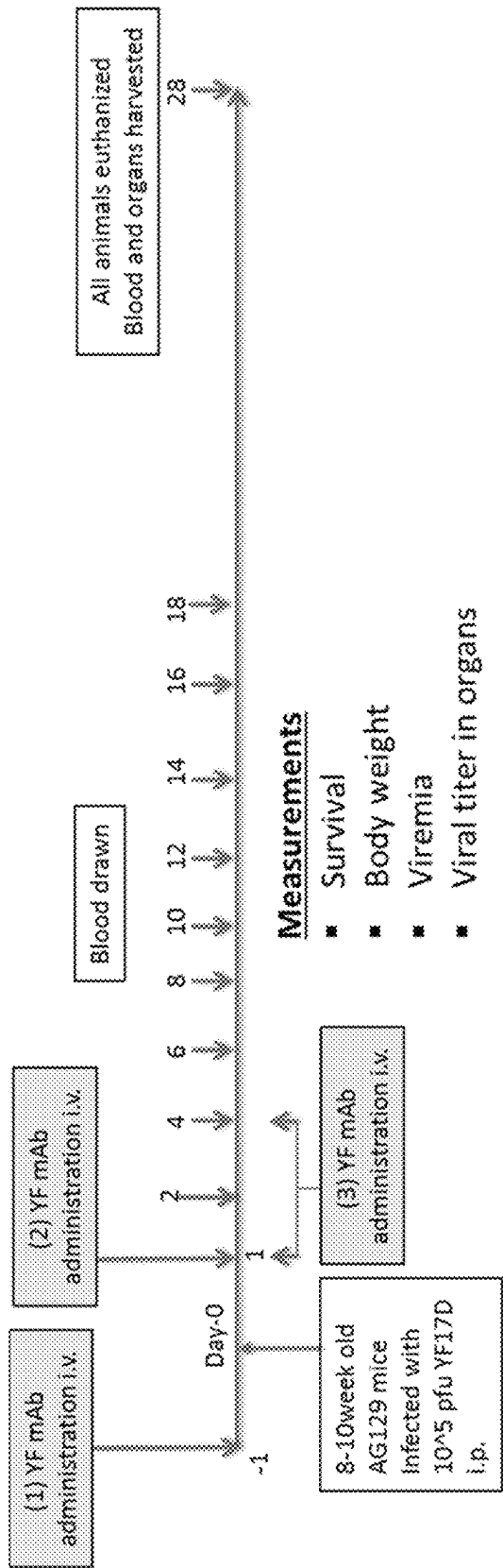


FIG. 1

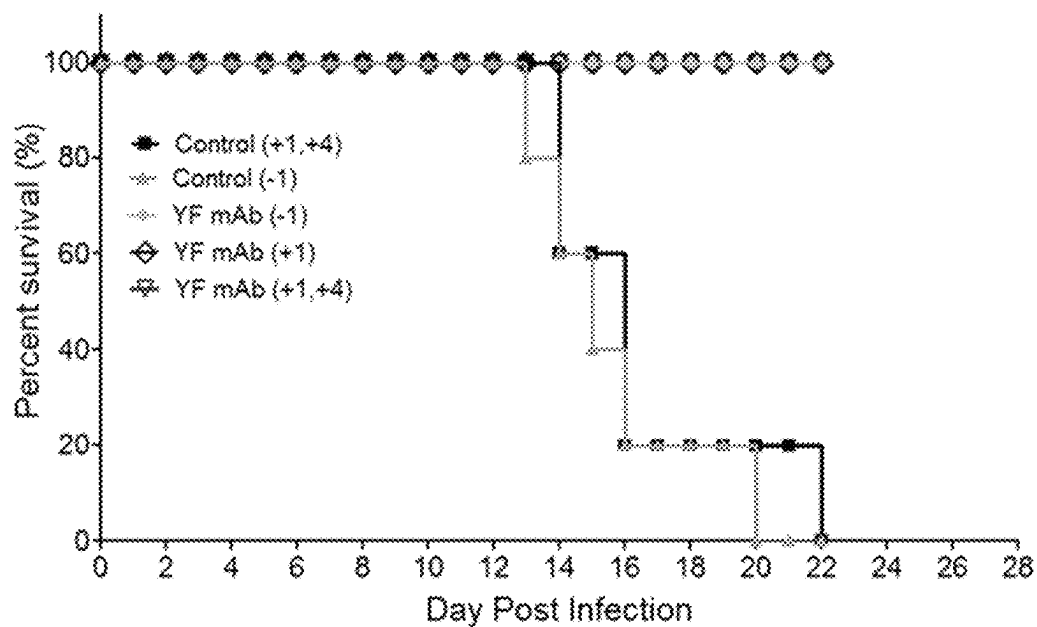


FIG. 2A

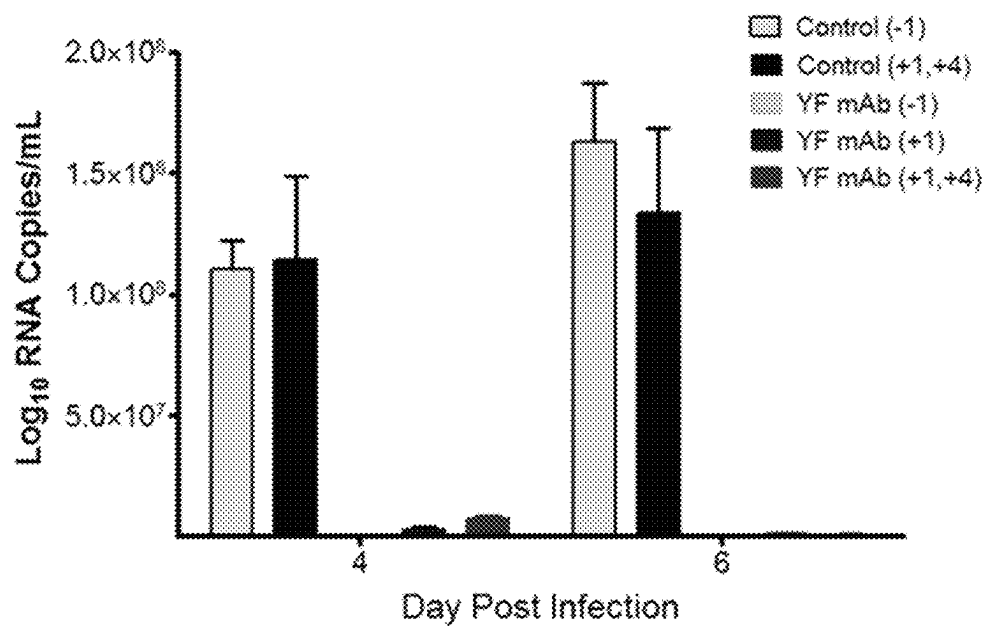


FIG. 2B

METHODS AND COMPOSITIONS FOR TREATING YELLOW FEVER

RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/US2019/027074, filed Apr. 11, 2019, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application No. 62/656,352, filed Apr. 11, 2018, the entire contents of each of which are incorporated herein by reference.

BACKGROUND

Yellow fever is an acute viral haemorrhagic disease caused by the Yellow Fever Virus (YFV). In humans the primary vector for YFV transmission is the mosquito *Aedes aegypti*, which transmits several other viruses including Dengue and Zika Virus. YFV is endemic in tropical and subtropical areas of Africa and Central and South America. Infected individuals develop a wide range of symptoms from asymptomatic infection to acute viral hemorrhagic disease (10-15%), of which there is a 50% fatality rate. Since the 1930s, a live attenuated vaccine has been available on the market. However, a global shortage in supplies have hampered efforts to prevent and control YFV outbreaks. The most recent indication of this can be seen in Brazil, where the YFV has already infected over 700 people and claimed more than 200 lives since the outbreak began in 2017. Currently, there is no YFV therapy available to treat those who have not been vaccinated or in which the vaccine has not provided a protective effect. There is a need for YFV therapies.

SUMMARY OF THE INVENTION

The present disclosure is based, at least in part, on the discovery of antibodies that specifically bind YFV and/or neutralize the virus. For example, the antibodies, or antigen-binding portions thereof, can prevent YFV from infecting cells in some embodiments. In an embodiment, the antibodies, or antigen-binding portions thereof, of any one of the compositions or methods provided herein, specifically bind to engage the key epitope residues (N106, K93 and K104) that are the most solvent exposed and antibody-accessible on the E-protein. In an embodiment, the antibodies, or antigen-binding portions thereof, of any one of the compositions or methods provided herein, have energetically favorable paratope (CDR) interactions around these key epitope residues and/or are characterized by a threshold minimum binding energy.

Accordingly, one aspect of the present disclosure provides an antibody or antigen-binding portion thereof that specifically binds to the E-protein or an Envelope Protein Domain II (E-DII) epitope of Yellow Fever Virus. In an embodiment of any one of the compositions or methods provided herein, the E-protein or E-DII epitope comprises an asparagine at position 106, a lysine at position 93, and a lysine at position 104 of Yellow Fever Virus E-protein.

In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion comprises the three CDRs of a heavy chain variable region (VH). In an embodiment of any one of the methods or compositions provided, the three CDRs are those found in any one of the VH sequences set forth herein, such as in Table 1 (e.g., SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,

28, 29, 30, 31, 32, 33, 34, 35, and 36). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises the three CDRs of a light chain variable region (VL). In an embodiment of any one of the methods or compositions provided, the three CDRs are those found in any one of the VL sequences set forth herein, such as in Table 2 (SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion comprises the three CDRs of any one of the VH sequences set forth herein, such as in Table 1 (e.g., SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36) and the three CDRs of any one of the VL sequences set forth herein, such as in Table 2 (SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises the three CDRs of the VH sequence and the three CDRs of the VL sequence of any one of the specific combinations of VH sequences and VL sequences as set forth in Table 3. Thus, in an embodiment of any one of the methods or compositions provided herein, the antigen-binding portion thereof is an antigen-binding portion of such an antibody.

Provided herein in one aspect is a nucleic acid encoding the three CDRs of any one of the VH sequences provided herein, such as provided directly above, and/or the three CDRs of any one of the VL sequences provided herein, such as provided directly above.

In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises any one of the VH amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises any one of the VL amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises any one of the VH amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36 and any one of the VL amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises any one specific combination of the combinations of VH amino acid sequences and VL amino acid sequences provided herein, such as set forth in Table 3.

Provided herein in one aspect is a nucleic acid encoding any one of the VH sequences provided herein, such as provided directly above, and/or any one of the VL sequences provided herein, such as provided directly above.

In an embodiment of any one of the methods or compositions provided herein, the antibody or antigen-binding

portion thereof comprises three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to the three CDRs of any one of the VH sequences set forth herein, such as in Table 1 (e.g., SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to the three CDRs of any one of the VL sequences set forth herein, such as in Table 2 (SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion comprises three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VH sequences set forth herein, such as in Table 1 (e.g., SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36) and three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VL sequences set forth herein, such as in Table 2 (SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73). In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to the VH sequence and three CDRs that have at least 90%, 95%, 96%, 97%, 98% or 99% identity to the VL sequence of any one of the specific combinations of VH sequences and VL sequences provided herein, such as set forth in Table 3. Thus, in an embodiment of any one of the methods or compositions provided herein, the antigen-binding portion thereof is an antigen-binding portion of such an antibody.

Provided herein in one aspect is a nucleic acid encoding the three CDRs of any one of the VH sequences provided herein, such as provided directly above, and/or the three CDRs of any one of the VL sequences provided herein, such as provided directly above.

In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises a VH amino acid sequence that has at least 85%, 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VH amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises a VL sequence that has at least 85%, 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VL amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises a VH amino acid sequence that has at least 85%, 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VH amino acid sequences provided herein, such as set forth in any one of SEQ ID. NOs: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, and 36 and a VL sequence that has at least 85%, 90%, 95%, 96%, 97%, 98% or 99% identity to any one of the VL amino acid sequences provided herein, such as set forth in

any one of SEQ ID. NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, and 73. In an embodiment of any one of the methods or compositions provided, the antibody or antigen-binding portion thereof comprises a VH sequence and a VL sequence that each independently have at least 85%, 90%, 95%, 96%, 97%, 98% or 99% identity to the VH and VL sequences of any one specific combination of the combinations of VH amino acid sequences and VL amino acid sequences provided herein, such as set forth in Table 3.

Provided herein in one aspect is a nucleic acid encoding the VH sequence as provided directly above and/or the VL sequences as provided directly above.

In an embodiment of any one of the methods or compositions provided herein, the antigen-binding portion thereof is an antigen-binding portion of any one of the antibodies provided herein.

Any one of the antibodies described herein can be a full-length antibody. The antibody or antigen-binding portion thereof can be human, humanized, or chimeric in an embodiment of any one of the methods or compositions provided herein. The antibody or antigen-binding portion thereof can be a single-chain antibody in an embodiment of any one of the methods or compositions provided herein. Any of the antibodies described herein can be either monoclonal or polyclonal. These two terms do not limit the source of an antibody or the manner in which it is made. The antibody or antigen-binding portion thereof can be a monoclonal antibody or antigen-binding portion thereof in an embodiment of any one of the methods or compositions provided herein. Further, the antigen-binding portion thereof can be an scFv, a F(ab')₂ fragment, a dAb fragment, a Fab fragment, a Fab' fragment, an Fv, or a disulfide-linked Fv fragment in an embodiment of any one of the methods or compositions provided herein. The antibody or antigen-binding portion thereof may be a single domain antibody, a diabody, a multispecific antibody, a bispecific antibody, or a dual-specific antibody. The antibody or antigen-binding portions thereof may be an isolated antibody or antigen-binding portion thereof in an embodiment of any one of the methods or compositions provided herein.

Further disclosed are methods of treating Yellow Fever, or a disease or condition associated with Yellow Fever Virus, by administering a therapeutically effective amount of one or more antibodies or antigen-binding portions thereof that specifically bind YFV, to a subject in need of such treatment. In an embodiment of any one of the methods provided herein, the subject has not been vaccinated for the YFV or in which a YFV vaccination did not provide an adequate protective effect. In an embodiment of any one of the method or compositions provided herein, the antibodies or antigen-binding portions thereof are any one or more of the antibodies or antigen-binding portions thereof described herein. In an embodiment of any one of the methods or compositions provided herein, the antibody or antigen-binding portion thereof specifically binds to the E-protein or an E-DII epitope of YFV. In an embodiment of any one of the methods or composition provided herein, the antibody or antigen-binding portion thereof specifically binds to residues 93, 104, and 106 of the E-protein or E-DII epitope of Yellow Fever Virus E-protein. In an embodiment of any one of the methods or compositions provided herein, the amount of the antibody or antigen-binding portion thereof is effective in reducing one or more symptoms of Yellow Fever in the subject. Any one of the anti-YFV antibodies or antigen-binding portions thereof may be administered systemically,

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e.g., via an enteral route or via a parenteral route, in any one of the methods provided herein.

The subject to be treated in any one of the methods described herein can be a patient (e.g., a human patient) who has or is suspected of having Yellow Fever, or a disease or condition associated with Yellow Fever Virus. In an embodiment of any one of the methods provided herein, the subject is a human patient who has or is suspected of having acute viral hemorrhagic disease.

Also provided herein in some aspects are (a) pharmaceutical compositions for use in treating Yellow Fever or a disease or condition associated with Yellow Fever Virus (e.g., acute viral hemorrhagic disease) in a subject, the pharmaceutical composition comprising any one or more of the antibodies or antigen-binding portions thereof described herein and a pharmaceutically acceptable carrier; and (b) uses of the just-described antibodies or antigen-binding portions thereof in medicaments and/or in the manufacturing of a medicament for treatment of Yellow Fever or a disease or condition associated with Yellow Fever Virus in a subject.

Also provided herein in some aspects are methods for producing the antibodies or antigen-binding portions thereof, nucleic acids encoding any one of the antibodies or antigen-binding portions thereof, vectors that can comprise any one or more of the nucleic acids provided herein, and related host cells.

In one aspect the method for producing the antibodies or antigen-binding portions thereof is any one of the methods described herein. In one embodiment, the method comprises considering the distinct domain proximal structural regions present on the viral assembly (e.g. inter-chain interfaces near the icosahedral axes of symmetry) and selecting promising Fv scaffolds. Conventional epitope prediction methods use domain structures to predict epitope surface regions embedded within the domain region. However, neutralizing flaviviral antibodies can recognize quaternary epitope surfaces (spanning two or more E protein chains). Thus, traditional methods do not incorporate valuable information.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows an example an in vivo study design of efficacy of engineered mAbs against YF-17D-204 in a mouse model of infection. Efficacy of designed mAbs was tested in a lethal model of Yellow fever infection in AG129 mice. Protective efficacy of mAbs was tested in prophylaxis or as therapy.

FIGS. 2A-2B shows in vivo efficacy of designed mAb against Yellow Fever Virus. FIG. 2A shows the survival curve of YF-17D infected AG129 mice treated with mAb at a dose of 10 mg/kg as prophylaxis (-1) or as Therapy (+1 and +1, +4). FIG. 2B shows blood viral titers of yellow fever virus in treated animals compared with control groups at Days 4 and 6 post infection. The administration of mAb resulted in complete protection compared to control groups (FIG. 2A). Administration of mAb also led to greater than 2 Log 10 reduction in viremia at days 4 and 6 post infection (FIG. 2B).

DETAILED DESCRIPTION

The following description is merely intended to illustrate various embodiments of the invention. As such, specific embodiments discussed herein are not to be construed as limitations to the scope of the invention. It will be apparent

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to one skilled in the art that various changes or equivalents may be made without departing from the scope of the invention.

Various aspects of the disclosure relate to antibodies, antigen-binding portions thereof, and pharmaceutical compositions thereof, as well as nucleic acids, recombinant expression vectors and host cells for making such antibodies and fragments. Provided herein are antibodies and/or antigen-binding portions thereof that specifically bind Yellow Fever Virus (YFV) and/or neutralize the virus. In some embodiments, the antibody or antigen-binding portion thereof binds to an E-DII epitope of the Yellow Fever Virus and prevents the virus from infecting cells. In some embodiments, the antibody or antigen-binding portion thereof binds to an E-DII epitope comprising an asparagine at position 106, a lysine at position 93 and/or a lysine at position 104 of Yellow Fever Virus E-protein. The antibodies and antigen-binding portions, as provided herein, in some embodiments, are used to treat or prevent Yellow Fever Virus infection in a subject.

An antibody (interchangeably used in plural form), as used herein, broadly refers to an immunoglobulin (Ig) molecule or any functional mutant, variant, or derivation thereof. It is desired that functional mutants, variants, and derivations thereof, as well as antigen-binding portions, retain the essential epitope binding features of an Ig molecule.

Antibodies are capable of specific binding to a target through at least one antigen recognition site, located in the variable region of the immunoglobulin molecule. Generally, an intact or full-length antibody comprises two heavy chains and two light chains. Each heavy chain contains a heavy chain variable region (VH) and a first, second and third constant regions (C_{H1} , C_{H2} and C_{H3}). Each light chain contains a light chain variable region (VL) and a constant region (CL). The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. A full-length antibody can be an antibody of any class, such as IgD, IgE, IgG, IgA, or IgM (or sub-class thereof), and the antibody need not be of any particular class. Depending on the antibody amino acid sequence of the constant domain of its heavy chains, immunoglobulins can be assigned to different classes. There are five major classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, and several of these may be further divided into subclasses (isotypes), e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2. The heavy-chain constant domains that correspond to the different classes of immunoglobulins are called alpha, delta, epsilon, gamma, and mu, respectively. The subunit structures and three-dimensional configurations of different classes of immunoglobulins are well known.

The term "antigen-binding portion" refer to a portion or region of an intact or full-length antibody molecule that can bind specifically to a target. Preferably, antigen-binding portions provided herein retain the ability to specifically bind to YFV. An antigen-binding portion may comprise the heavy chain variable region (VH), the light chain variable region (VL), or both. Each of the VH and VL typically contains three complementarity determining regions CDR1, CDR2, and CDR3.

Examples of antigen-binding portions include, but are not limited to: (1) an Fab fragment, which can be a monovalent

fragment having a VL-CL chain and a VH-CH chain; (2) an F(ab')₂ fragment, which can be a bivalent fragment having two Fab fragments linked by a disulfide bridge at the hinge region, i.e. a dimer of Fab; (3) an Fv fragment having the VL and VH domains of a single arm of an antibody; (4) a single chain Fv (scFv), which can be a single polypeptide chain composed of a VH domain and a VL domain through a peptide linker; (5) a (scFv)₂, which can comprise two VH domains linked by a peptide linker and two VL domains, which are associated with the two VH domains via disulfide bridges; (6) a Fd fragment consisting of the VH and CH1 domains; (7) a dAb fragment, which comprises a single variable domain; and (8) an isolated complementarity determining region (CDR).

Furthermore, although the two domains of the Fv fragment, VL and VH, can be coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the VL and VH regions pair to form monovalent molecules (known as single chain Fv (scFv); see e.g., Bird et al. (1988) *Science* 242:423-426; and Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883). Such single chain antibodies are also intended to be encompassed within the term "antigen-binding portion" of an antibody. Other forms of single chain antibodies, such as diabodies are also encompassed. Diabodies are bivalent, bispecific antibodies in which VH and VL domains are expressed on a single polypeptide chain, but using a linker that is too short to allow for pairing between the two domains on the same chain, thereby forcing the domains to pair with complementary domains of another chain and creating two antigen binding sites (see e.g., Holliger, P., et al. (1993) *Proc. Natl. Acad. Sci. USA* 90:6444-6448; Poljak, R. J., et al. (1994) *Structure* 2: 1121-1123). Diabodies are also encompassed within the term "antigen-binding portion".

The term "human antibody" refers to antibodies having variable and constant regions corresponding substantially to, or derived from, antibodies obtained from human subjects, e.g., encoded by human germline immunoglobulin sequences or variants thereof. The human antibodies described herein may include one or more amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo). Such mutations may present in one or more of the

CDRs, particularly CDR3, or in one or more of the framework regions. In some embodiments, the human antibodies may have at least one, two, three, four, five, or more positions replaced with an amino acid residue that is not encoded by the human germline immunoglobulin sequence. However, the term "human antibody", as used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

The term "recombinant human antibody", as used herein, is intended to include all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as antibodies expressed using a recombinant expression vector transfected into a host cell, antibodies isolated from a recombinant, combinatorial human antibody library (Hoogenboom H. R., (1997) *TIB Tech.* 15:62-70; Azzazy H., and Highsmith W. E., (2002) *Clin. Biochem.* 35:425-445; Gavidondo J. V., and Larrick J. W. (2002) *BioTechniques* 29: 128-145; Hoogenboom H., and Chames P. (2000) *Immunology Today* 21:371-378), antibodies isolated from an animal (e.g., a mouse) that is transgenic for human immunoglobulin genes (see e.g., Taylor, L. D., et al. (1992) *Nucl. Acids Res.* 20:6287-6295; Kellermann S.-A., and Green L. L. (2002) *Current Opinion in Biotechnology* 13:593-597; Little M., et al (2000) *Immunology Today* 21:364-370) or antibodies prepared, expressed, created or isolated by any other means that involves splicing of human immunoglobulin gene sequences to other DNA sequences. Such recombinant human antibodies have variable and constant regions as defined above. In certain embodiments, however, such recombinant human antibodies may be subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the VH and VL regions of the recombinant antibodies may be sequences that, while derived from and related to human germline VH and VL sequences, may not naturally exist within the human antibody germline repertoire in vivo.

Some embodiments of the disclosure provide fully human antibodies capable of binding the E-DII epitope of Yellow Fever Virus. In some embodiments, the E-DII epitope comprises an asparagine at position 106, a lysine at position 93, and a lysine at position 104 of Yellow Fever Virus E-protein.

The protein sequences for the various VH and VL regions are displayed in Table 1 and Table 2 respectively.

TABLE 1

Protein sequence for variable heavy chains

```
>VH.01; SEQ ID NO: 1
EVKLVESGGGLVKGSSSLKLSAASGFTFTNYAMSWVRQTPEKRLEWVASISSGHTPPYPDSVKGRFTISRDN
RNILFLQMSSLRSEDAMYYCARGDYYGVSYSAMDYWGQGTSTVTVSS

>VH.11; SEQ ID NO: 2
EVQLVESGGGLVQPGRSLRLSCAASGFTFTNYAMSWVRQAPGKLEWVSSISSGHTPPYPDSVKGRFTISRDN
KKSLLYLMNSLRAEDTALYYCARGDYYGVSYSAMDYWGQGTSTVTVSS

>VH.21; SEQ ID NO: 3
EVQLVESGGGLVQPGRSLRLSCAASGFTFTDYYMSWVRQAPGKLEWVSSISSGHTPPYPDSVKDRFTISRDN
KKSLLYLMNSLRAEDTALYYCARGDYYGTVYSAMDYWGQGTSTVTVSS

>VH.31; SEQ ID NO: 4
EVQLVESGGGLVQPGRSLRLSCAASGYAFTNYAMSWVRQAPGKLEWVSSISSGHTPPYPDTVKGRFTISRDN
AKKSLLYLMNSLRAEDTALYYCARGDYYGSSYSAMDYWGQGTSTVTVSS

>VH.3.11; SEQ ID NO: 5
EVQLVESGGGLVQPGRSLRLSCAASGYAFTNYGVNWRQAPGKLEWVSSISSGGSTYPDSVKGRFTISRDN
KKSLLYLMNSLRAEDTALYYCARHDYYGSSY-AMDYWGQGTSTVTVSS
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TABLE 1-continued

Protein sequence for variable heavy chains
>VH.3.21; SEQ ID NO: 6 EVQLVESGGGLVQPGRSLRLSCAASGYAFTNYGVNWRQAPGKGLEWVSSISSGGSTYYPD SVKGRFTISRDN KKSLEYLQMNSLRAEDTALYYCARSHYYGSSYDAMDYWGQGT TTVTVSS
>VH.41; SEQ ID NO: 7 EVQLVESGAEVKKPGSSVKVSCASGFTFTNYAMSWVRQAPGQGP EWMGSISSGHTPYYPDSVKGRVTITA DDFAGTVYMELSSLRSED TAMYCRGDYYGSVYSAMDYWGKGT TTVTVSS
>VH.51; SEQ ID NO: 8 EVQLVESGAEVKKPGSSVKVSCASGFTFTNYAMSWVRQAPGQGP EWMGSISSGHTPYYPDSVKGRVTITAD DFAGTVYMELSSLRSED TAMYCARGDYYSVYSAMDYWGKGT TTVTVSS
>VH.61; SEQ ID NO: 9 QVQLVQSGAEVKKPGASVKVSCAGFTFTNYAMSWVRQAPGQGLEWMGSISSGHTPYYPDSVKGRVTMTAD TSTNTAYMELRSLRSD TAVYYCARGDYYSVYSAMDYWGQGT TTVTVSS
>VH.71; SEQ ID NO: 10 EVQLVESGGGLVQPGGSLRLSCASGFTFTNYAMSWVRQAPGKGLEWVSSISSGHTPYYPDSVKGRFTISRDN SK NTLYFEMNSLRPED TAVYYCVRGDYYSVYSAMDYWGQGT TTVTVSS
>VH.1; SEQ ID NO: 11 ASVLSEVQLQQSGPELVKPGASVKLSCKTSENTFT EYTMHWVKQSHGKSL EWIGGIDPNNGGTNYNQFKGK ATLTVDKSSNTAYMELRSLTSEDSAVYYCGRRDY YALDYWGQGT SVTVAS
>VH.2; SEQ ID NO: 12 QVQLQQPGAEVLKPGASVKLSCKASGYTFTSYW MHVVKLRPGQGF EWIGDINPNNGGPSYNEKFKRKATLT V DTSSNTAYMQLSSLTSEDSAVYYCTID DGYRFGYWGQGT LTVTSA
>VH.3; SEQ ID NO: 13 QVQLQQSGSEL MKPGASVQISCKATGYTFS D YWIEWVKQRP GHGLEWIGDILCGTGRTRYNEKLKAMATFTA DTSSNTAFMQLSSLTSEDSAVYYCARSASYG DYADYWGHTT LTVSS
>VH.4; SEQ ID NO: 14 AQLQQSGTGLARPGASVKLSCKASGYTFTSYGISWVTQRAGQGLEWIGVIYPRSGNTYYNEKFRGKATLTADKS SSSAYMELRGLTAEDSAVYFCARENYGSVYWGQGT TLTVSS
>VH.5; SEQ ID NO: 15 EVQLQQSGAEVLKPGASVKLSCTASGFNIKDTYMHVVKQRPEQGLEWIGRIDPANGYSKYDPKFKGKATITAD TSSNAAYLQLSSLTSED TAVYFCARDYEGFAYWGQGT LTVTVSS
>VH.6; SEQ ID NO: 16 EVQLVESGGGVQPGRSLRLSCAASGFSFSSYGMHWRQAPGKGLEWVAVIWDYDGS KTYYGDSVKGRFTISK DNSKKMVLQMDSLGVEDTAFYYCARGIAGGWAFW GIDLWGQGT LTVTVSS
>VH.7; SEQ ID NO: 17 EVQLQQSGAEVLKPGASVKLSCTASGFNIKDTYMHVVKQRPEQGLEWIGRIDPANGYSKYDPKFKGKATITAD TSSNAAYLQLSSLTSED TAVYFCARDYEGFAYWGQGT LTVTVSS
>VH.8; SEQ ID NO: 18 EVQLVQSGAEVRKPGASTKVSCASGYTFTHYMHVWRQAPGQGLEWMGIINPSGGSTTYAQLKQGRVTMT RDTSTSTVYMELSSLRSED TAVYYCARDWGSNYVWGSYPKYWGQGT LTVTVSS
>VH.9; SEQ ID NO: 19 EVQLVESGGGLVQPGRSLRLSCAASGFTFNDYAMHWRQAPGKGLEWVSTISWNSGSI GYADSVKGRFTISR DNAKKSLEYLQMNSLRAEDTALYYCAKDIQYGNYYYGMDVWGQGT TTVTVSS
>VH.10; SEQ ID NO: 20 DVQLVEPGAEVLQPGASVKMSCKASGYTFSSYWINWEKQRP GKGLEWIGNIYPGSGTVNYDDKFKSKATLTID TSSNTAYMQLSSLTSEDSAVYYCTRGSHAMDYWGQGT SVTVSS
>VH.11; SEQ ID NO: 21 EVQLVESGGGLVRPGGSLRLSCAASGFSYSNHHMHWVRQAPGKGLVWVSRINSDGSTRNYADFVKGRFTISR DNAENTLYLEMNLSLTAD TAVYYCVRDGVRFYYDSTGYYPDSFFKYGMDVWGQGT TTVTVSS
>VH.12; SEQ ID NO: 22 EVQLVESGGGLVRPGGSLRLSCAASGFSYSNHHMHWVRQAPGKGLVWVSRINSDGSTRNYADFVKGRFTISR DNAENTLYLEMNLSLTAD TAVYYCVRDGVRFYYDSTGYYPDSFFKYGMDVWGQGT TTVTVSS
>VH.13; SEQ ID NO: 23 EVKLVESGGGLVLPGGSLRLSCATSGFTFTDYMTWVRQPPGKALEWLGFIGNKANDYTTEYSASVKGRFTISR DDSQSILY LQMS TLRAEDRATYYCATVYGNYPYFDVWGAGTT VAVSS
>VH.14; SEQ ID NO: 24 EVQLVESGAEVKKPGSSVKVSCASGGTFNNYAI SWVRQAPGQGLEWMGGIIPFGGANYAQKFQGRVTITAD RSTSTVYMELSGLRSED TAVYYCARRPQSI FDNFDFLWGRGTLTVTVSSAGTKGPS

TABLE 1-continued

Protein sequence for variable heavy chains
>VH.15; SEQ ID NO: 25 EVQLVESGGGLVQPGGSLKLSCAASGFTFSSHHMHWVRQAPGKGLVWVSRITNSDGSSTSYADSVKGRFMIS RDNSKNTVYLHMGRLRAEDTAVYFCARDGVRIYYDSTGYYPDNPFGYGLDVWGQGTITVTVSSA >VH.16; SEQ ID NO: 26 EVQLQQSGPELVKPGASVKISCKASGYTFTDYNMHWVKQSHGKSLIEWVGYTYPYNGGIGYNQKFKSKATLTLD NSSRTAYMELRSLTSEDSAVYYCVRRGRIYDGAHFDYWGQGTITLTVSS >VH.17; SEQ ID NO: 27 DVQLVESGGGLVQPGGSRKLSCAASGFTFSSFGMHWVRQAPKGLIEWVAYISSGSSTLHYADTVKGRFTISR NPKNTLFLQMTSLRSEDATAMYYCARWGNYPHYAMDYWGQGTITVTVSS >VH.18; SEQ ID NO: 28 QVQLVQSGAEVKKPGASVKVSKAGFNIDVYMSWVRQAPKGLIEWMGRIDPENGDTKYDPKLQGRVTMT ADTSTNTAYMELRSLRSDDTAVYYCARGWEGFAYWGQGTITVTVSS >VH.19; SEQ ID NO: 29 EVQLVESGGGLVQPGGSLRLSCASGFTFSTYSMHVVRQAPGKGLIEYVSAITGEGDSAFYADSVKGRFTISRDN SKNTLYFEMNSLRPEDTAVYYCVGGYSNFIYYTMDVWGQGTITVTVSSG >VH.20; SEQ ID NO: 30 EVQLVESGGGLVQPGGSLRLSCASGFSYSNHHMHWVRQAPGKGLVWVSRINSDGSTRNYADTVKGRFTISR DNAENTLYLEMNSLTADDTAVYYCVRDGVRFYDSTGYYPDSFFKYGMVDVWGQGTITVTVSS >VH.21; SEQ ID NO: 31 DVQLVESGGGLVQPGGSRKLSCAASGFTFSSFGMHWVRQAPKGLIEWVAYISSGSSTLHYADTVKGRFTISR NPKNTLFLQMTSLRSEDATAMYYCARWGNYPHYAMDYWGQGTITVTVSS >VH.22; SEQ ID NO: 32 QVQLQQPGAEVLKPGASVKMSCKASGYTFTSYNMHWVKQTPGRGLEWIGAIYPNGDTSYNQKFKGKATLT ADKSSSTAYMQLSSLTSEDSAVYYCARSTYYGGDWYFNVWGAGTITVTVSA VH.23; SEQ ID NO: 33 QVQLVQSGAEVKKPGSSVKVSKASGYAFSYSWINWVRQAPKGLIEWMGRIFPGDGDYNGKFKGRVTIT ADKSTSTAYMELSSLRSEDATAVYYCARNVFDGYWLVIYWGQGTITVTVSS >VH.24; SEQ ID NO: 34 EVQLVESGAEVKKPGSSVKVSKASGGPFRSYAISWVRQAPKGLIEWMGGIIPFGTITKYAPKFKGRVTITADD FAGTVYMESSLRSEDATAMYYCAKHMGYQVRETMVWGKGTITVTVSS >VH.25; SEQ ID NO: 35 EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYIMMWVRQAPKGLIEWVSSIYPSGGITFYADTVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARIKLGTVTTVDYWGQGTITVTVSS >VH.26; SEQ ID NO: 36 QVTLRESGPALVKPTQTLTLCTFSGFSLSTAGMSVGVIRQPPGKALEWLADIWDDKKHYNPSLKDRLTISKD TSKNQVVLKVTNMDPADTATYYCARDMIFNFYFDVWGQGTITVTVSS

TABLE 2

Protein sequence for variable light chains
>VL.0; SEQ ID NO: 37 DIRMTQSPSSMYASLGERVTITCKASQDINSYLSWLQQKPGKSPKTLIYRANRLFDGVPSRFSGSGSQDYSLTIS SLEYEDMGIFYCLQYDEFPFTFGSGTKLEIK >VL.111; SEQ ID NO: 38 DIRMTQSPSSLSASVGRVTITCKASQDINSYLSWLQQKPGKSPKTLIYRANRLFDGVPSRFSGSGSGTDFTLTISS LQPEDFAIYYCLQYDEFPFTFGSGTKVEIK >VL.1.11; SEQ ID NO: 39 DIRMTQSPSSLSASVGRVTITCKASQDINSYLNWLQQKPGKSPKTLIYRVNRLVDGVPSRFSGSGSGTDFTLTIS SLQPEDFAIYYCLQYDEFPYTFGSGTKVEIK >VL.1.21; SEQ ID NO: 40 DIRMTQSPSSLSASVGRVTITCKASQDIKSYLSWLQQKPGKSPKTLIYRVNRLVDGVPSRFSGSGSGTDFTLTISS LQPEDFAIYYCLQYDEFPYTFGSGTKVEIK >VL.1.31; SEQ ID NO: 41 DIRMTQSPSSLSASVGRVTITCKASQDINSYLNWLQQKPGKSPKTLIYRVNRLVDGVPSRFSGSGSGTDFTLTIS SLQPEDFAIYYCLHYDEFPYTFGSGTKVEIK >VL.1.41; SEQ ID NO: 42 DIRMTQSPSSLSASVGRVTITCKASQDIKSYLSWLQQKPGKSPKTLIYRVNRLVDGVPSRFSGSGSGTDFTLTISS LQPEDFAIYYCLHYDEFPYTFGSGTKVEIK

TABLE 2-continued

Protein sequence for variable light chains

>VL.211; SEQ ID NO: 43
DIRMTQSPSSLSASVGDRVITICRASQDINSYLSWLQQKPGKSPKTLIYRANRLMIGVPSRFSGSGSGTDFTLTIS
LQPEDFAIYYCLQYDDFPLTFGSGTKVEIK

>VL.311; SEQ ID NO: 44
DIRMTQSPSSLSASVGDRVITICKASQDINSYLSWLQQKPGKSPKTLIYRANRVFDGVPSRFSGSGSGTDFTLTIS
LQPEDFAIYYCLQYDDFPLTFGSGTKVEIK

>VL.411; SEQ ID NO: 45
QSVLTQPPSVSAAPGQKVTISCKASQDINSYLSWYQQLPGTAPKLLIYRANRLFDGIPDRFSGSKSGTSATLGITGL
QTGDEANYCYLQYDEFPFTFGGKTLTVL

>VL.511; SEQ ID NO: 46
DIVMTQSPASLAVSLGQRATISCKASQDINSYLSWYQQKPGQPPKLLIYRANRLFDGVPDRFSGSGSGTDFTLTIS
SLQAEADVAVYYCLQYDEFPFTFGGKLEIKR

>VL.611; SEQ ID NO: 47
EIVLTQSPATLSLSPGERATLSCKASQDINSYLSWYQHKPGQAPRLLIYRANRLFDGVPARFSGSRSGTDFTLTISTL
EPEDFAVYYCQLQYDEFPFTFGGKVEIK

>VL.1; SEQ ID NO: 48
DIVMTQSQKFMSTSVGDRVSIICKASQHVGSAAVWYQQKPGQSPKLLIHSASNRYTGVPDRFTGSGSGTDFTLT
ISNIQSEDLADYFCQQYNSYPTFGGKLEIK

>VL.2; SEQ ID NO: 49
DIQMTQSPASLSASVGETVTITCRASGNIHNLAWYQQKQKSPQLLVYNAKTLADGVPSRFSASGSGTQYSLKI
NSLQPEDFGSYCYQHFWSTPRTFGGGKLEIKR

>VL.3; SEQ ID NO: 50
DIVMTQSHKFMSTSVGDRVSIICKASQDVSTAVAWYQQKPGQSPKLLISWASTRHTGVPDRFTGSGSGTDYTLT
ISSVQAEDLALYYCQHYTTPFTFGAGTKLELK

>VL.4; SEQ ID NO: 51
DIVMTQSQKFMSTSVGDRVSIICKASQNVGTAVAWYQQKPGQSPKLLIYSASNRYTGVPDRFTGSGSGTDFTLT
ISNMQSEDLADYFCQQFSSYPYTFGGGKLEIK

>VL.5; SEQ ID NO: 52
DIVLTQSPASLAVSLGQRATISCRASESVVRYGNSFMHWYQQKPGQPPKLLIYRASSLESIGIPTRFSGSGSRTDFTL
TINPVEADDVATYYCQQTNPDPWAFGGGKLEIK

>VL.6; SEQ ID NO: 53
DVVMTQSPGTLSSLSPGERATLSCRASQNVSYLWYQHKPGRSPRLIFGVTSRATGVPDRFSGSGSGTDFTLTIS
RLEPEDFAVYYCQQYAGSAYTFGGGKVEIKR

>VL.7; SEQ ID NO: 54
DIVLTQSPASLAVSLGQRATISCRASESVVRYGNSFMHWYQQKPGQPPKLLIYRASSLESIGIPTRFSGSGSRTDFTL
TINPVEADDVATYYCQQTNPDPWAFGGGKLEIK

>VL.8; SEQ ID NO: 55
QSVLTQPSVSGTGPQRTISCSGSSNIGSNTVNWYQQLPGTAPKLLIYGNQRPSPGVDRFSGSKSGTSASLAI
SGLQSEDEADYYCAAWDDSLNGPVFGGKTLTVLG

>VL.9; SEQ ID NO: 56
DIQMTQSPSSLSASVGDRVITICRTSQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISL
QPEDFATYYCQSYSTPRTFGGGKVEIK

>VL.10; SEQ ID NO: 57
DIVMTQSQKFMSTSVGDRVSIICKASQNVRTSVAWYQQKPGQSPKALIYLASNRHTGVPDRFTGSGSGTDFTLT
ISNVQSEDLADYFCLQHWTPYPTFGGKLEIK

>VL.11; SEQ ID NO: 58
GASQSVLTQPVSVSGSPGQSIITISCTGTSSNADTYNLVSWYQQRPKGAPKLLIYEGTKRPSGVSNRFSASKSATA
ASLTISGLQPEDEADYYCCSYATSRTLVFGGKTLTVV

>VL.12; SEQ ID NO: 59
GASQSVLTQPVSVSGSPGQSIITISCTGTSSNADTYNLVSWYQQRPKGAPKLLIYEGTKRPSGVSNRFSASKSATA
ASLTISGLQPEDEADYYCCSYATSRTLVFGGKTLTVVAA

>VL.13; SEQ ID NO: 60
QIVLTQSPAISASPGKVTMTCSASSSVYMHWYQQKSGTSPKIWIYESSKLASGVPRFSGSGSGTSYSLTIS
MEAEDVATYYCQQWSSHPHPLTFGAGTKLELK

>VL.14; SEQ ID NO: 61
QSVLTQPPSASGTGPQRTISCSGSSNVGSNYVWYQQLPGTAPKLLIYRNNRRPSPGVDRFSGSKSGTSASLAI
SGLRSEDEADYYCATWDDSLSGLVFGGKTLTVLGQPKA

TABLE 2-continued

Protein sequence for variable light chains	
>VL.15; SEQ ID NO: 62	RSQSALTQPASVSGSPGQSITISCTGISSDVETYNLVSWEQHPGKAPKLIIEASKRPSGVSNRFSGSKSGNTASLAISGLQAEDEADYYCCSYAGGKSLVFGGGTRLTVLGQP
>VL.16; SEQ ID NO: 63	DIKMTQSPSSMYASLGERVTITCKASQGINSLSWFQKPGKSPKTLIYRANRLVDGVPSPRFSGSGSQDYSLTIS SLEYEDMGIYYCLQYDEFPLTFGAGTKLELK
>VL.17; SEQ ID NO: 64	NIVMTQSPKSMMSVGERVTLTCKASENVGTYSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTI SSVQAEDLADYHCGQSYSTPYTFGGGKLEIK
>VL.18; SEQ ID NO: 65	DIVMTQSPASLAVSLGQRATISCRASENVDKYNSFMHWYQQKPGQPPLLIYRASELQWGVDPDRFSGSGSGT DFTLTISLQAEADVAVYYCQRSNEVPWTFGGTKLEIKRTVAHHHHHH
>VL.19; SEQ ID NO: 66	SEIVLTQSPATLSLSPGERATLSCRASQSIISFLAWYQHPKQAPRLLIYDASTRATGVPARFSGSRSGTDFTLTISTL EPEDFAVYYCQQRYNWPPYTFGGTKVEIK
>VL.20; SEQ ID NO: 67	QSVLTQPVSVSGSPGQSITISCTGTSSNADTYNLVSWEQKPGKAPKLMIEGTRKPSGVSNRFSASKSATAASLT ISGLQPEDEADYYCCSYATSRTLVFGGGTKLTVVGQP
>VL.21; SEQ ID NO: 68	NIVMTQSPKSMMSVGERVTLTCKASENVVTYSWYQQKPEQSPKLLIYGASNRYTGVPDRFTGSGSATDFTLTI SSVQAEDLADYHCGQSYSTPYTFGGGKLEIK
>VL.22; SEQ ID NO: 69	QIVLSQSPAILSASPGKVTMTCRASSSVSYIHWFPQKPGSSPKPWIYATSNLASGVPVRFSGSGSGTSYSLTISRVAEADAATYYCQQTWTSNPPTFGGGKLEIK
>VL.23; SEQ ID NO: 70	DIVMTQTPLSLPVTPGEPASISCRSSKSLHNSGITYLYWYQKPGQSPQLLIYQMSNLVSGVPDRFSGSGSGTDFT LKISRVEAEDVGYYCAQMLELPYTFGGGKVEIK
>VL.24; SEQ ID NO: 71	QSVLTQPPSVSAAPGQKVTISCSGSSNIGNDYVSWYQQLPGTAPKLLIYDNKRPSGIPDRFSGSKSGTSATLGIT GLQTGDEANYYCATWDRRPTAYVVFSGGKLTVL
>VL.25; SEQ ID NO: 72	QSALTQPASVSGSPGQSITISCTGTSSDVGGYNYVSWYQHPGKAPKLMIDVSNRPSGVSNRFSGSKSGNTAS LTISGLQAEDEADYYCCSYTSSSTRVFGTGKVTVL
>VL.26; SEQ ID NO: 73	DIQMTQSPSTLSASVGRVTITCSASSRVGYMHYQKPGKAPKLLIYDTSKLAGVPSRFSGSGSGTEFTLTISSL QPDDFATYYCFQGSQYPTTFGGGKVEIK

An “isolated” substance means that it has been altered by the hand of man from the natural state. If an “isolated” substance presents in nature, it has been changed or removed from its original environment, or both. For example, a polypeptide naturally present in a living subject is not “isolated” but the polypeptide is isolated if it has been substantially separated from the coexisting materials of its natural state and/or exists in a substantially pure state.

The term “specifically binds” or “specifically binding” refers to a non-random binding reaction between two molecules, such as the binding of the antibody or antigen-binding portion thereof to an epitope of the antigen. An antibody or antigen-binding portion thereof that “specifically binds” to a target or an epitope is a term well understood in the art, and methods to determine such specific binding are also well known in the art. A molecule is said to exhibit “specific binding” if it reacts or associates more frequently, more rapidly, with greater duration and/or with greater affinity with a particular target antigen or an epitope than it does with alternative targets/epitopes. An antibody or antigen-binding portion thereof “specifically binds” to a target antigen if it binds with greater affinity, avidity, more readily, and/or with greater duration than it

binds to other substances. In certain embodiments, an antibody is said to specifically bind an antigen when it preferentially recognizes its target antigen in a complex mixture of proteins and/or macromolecules.

An “epitope” is a region of an antigen that is bound by an antibody. The term includes any polypeptide determinant capable of specific binding to an immunoglobulin. In certain embodiments, epitope determinants include chemically active surface groupings of molecules such as amino acids, sugar side chains, phosphoryl, or sulfonyl, and, in certain embodiments, may have specific three dimensional structural characteristics, and/or specific charge characteristics.

As used herein, the term “neutralizing” refers to neutralization of an activity, such as a biological activity, of a target protein (e.g., Yellow Fever Virus E-protein). In one embodiment, a neutralizing antibody binds to the E-DII epitope of Yellow Fever Virus E-protein and results in inhibition of a biological activity of Yellow Fever Virus and/or prevents the virus from infecting cells.

Subjects may be human, but also include other mammals, particularly those mammals useful as laboratory models for human disease, e.g. mouse, rat, rabbit, dog, etc.

The term “treat”, “treatment” or “treating” refers to an action, application or therapy, wherein a subject, including

a human being, is subjected to medical aid with the purpose of improving the subject's condition, directly or indirectly. Particularly, the term refers to reducing incidence, or alleviating one or more symptoms, eliminating recurrence, preventing recurrence, preventing incidence, improving one or more symptoms, and/or improving prognosis or a combination thereof in some embodiments. The skilled artisan would understand that treatment does not necessarily result in the complete absence or removal of symptoms. For example, with respect to Yellow Fever Virus, "treatment" or "treating" may refer to reducing the severity or duration of acute viral haemorrhagic disease caused by Yellow Fever.

An "effective amount" or an "effective dose" or a "therapeutically effective amount" in connection with administration of a pharmacological agent, as used herein, refers to an amount of a drug or pharmaceutical agent (e.g., an antibody or antigen-binding fragment or portion thereof, or a composition comprising the same) which, as compared to a corresponding subject who has not received such amount, results in an intended pharmacological result, or an effect in treatment, healing, prevention, or amelioration of a disease or disease symptom, disorder, or side effect, or a decrease in the rate of advancement of a disease or disorder, or any symptom thereof. The effective amount or dose of a pharmacological agent may vary depending on the particular active ingredient employed, the mode of administration, and/or the age, size, and condition of the subject to be treated.

The present disclosure also provides compositions (e.g., pharmaceutical compositions) comprising an antibody or antigen-binding portion thereof that binds specifically to the E-DII epitope of Yellow Fever Virus E-protein. In some embodiments, the E-DII epitope comprises an asparagine at position 106, a lysine at position 93, and a lysine at position 104 of Yellow Fever Virus E-protein. Compositions can be prepared from any one of the antibodies or antigen-binding portions described herein. The antibodies or antigen-binding portions thereof, as well as the encoding nucleic acids or nucleic acid sets, vectors comprising such, or host cells comprising the vectors, as described herein can be mixed with a pharmaceutically acceptable carrier (excipient), such as to form a pharmaceutical composition for use in treating a target disease. Pharmaceutically acceptable excipients (carriers) including buffers, are well known in the art. See, e.g., Remington: The Science and Practice of Pharmacy 20th Ed. (2000) Lippincott Williams and Wilkins, Ed. K. E. Hoover.

The pharmaceutical compositions can comprise pharmaceutically acceptable carriers, excipients, or stabilizers in the form of lyophilized formulations or aqueous solutions. (Remington: The Science and Practice of Pharmacy 20th Ed. (2000) Lippincott Williams and Wilkins, Ed. K. E. Hoover). Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations used, and may comprise buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including

glucose, mannose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g., Zn-protein complexes); and/or non-ionic surfactants such as TWEENTM (polysorbate), PLURONICSTM (poloxamers) or polyethylene glycol (PEG).

The phrase "pharmaceutically acceptable", as used in connection with compositions of the present disclosure, refers to molecular entities and other ingredients of such compositions that are physiologically tolerable and do not typically produce untoward reactions when administered to a subject. Preferably, as used herein, the term "pharmaceutically acceptable" means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in mammals, and more particularly in humans. "Acceptable" means that the carrier is compatible with the active ingredient of the composition (e.g., the nucleic acids, vectors, cells, or therapeutic antibodies) and does not negatively affect the subject to which the composition(s) are administered. Any of the pharmaceutical compositions to be used in the present methods can comprise pharmaceutically acceptable carriers, excipients, or stabilizers in the form of lyophilized formations or aqueous solutions.

Pharmaceutically acceptable carriers, including buffers, are well known in the art, and may comprise phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives; low molecular weight polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; amino acids; hydrophobic polymers; monosaccharides; disaccharides; and other carbohydrates; metal complexes; and/or non-ionic surfactants. See, e.g. Remington: The Science and Practice of Pharmacy 20th Ed. (2000) Lippincott Williams and Wilkins, Ed. K. E. Hoover.

A pharmaceutical composition can be presented in unit dosage form and can be prepared by any suitable method, many of which are well-known. Such methods can include the step of bringing an anti-YFV antibody into association with a carrier that constitutes one or more accessory ingredients.

Compositions provided herein can be a sterile aqueous preparation, which preferably in some embodiments is isotonic with the blood of the recipient. This aqueous preparation can be formulated according to known methods using suitable dispersing or wetting agents and suspending agents. The sterile aqueous preparations also can be a sterile injectable solution or suspension in a non-toxic parenterally-acceptable diluent or solvent. Among the acceptable vehicles and solvents that can be employed are water, Ringer's solution, and isotonic sodium chloride solution.

The term "vector", as used herein, is intended to refer to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a "plasmid", which refers to a circular double stranded DNA loop into which additional DNA segments may be ligated. Moreover, vectors may be capable of directing the expression of genes to which they are operatively linked. Such vectors are referred to herein as "recombinant expression vectors" or "expression vectors".

The term "recombinant host cell" or "host cell", as used herein, is intended to refer to a cell into which exogenous DNA has been introduced. It should be understood that such terms are intended to refer not only to the particular subject cell, but, to the progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny

may not, in fact, be identical to the parent cell, but are still included within the scope of the term “host cell” as used herein.

Examples

Anti-YFV monoclonal antibodies are designed using structure-guided analysis of the YFV envelope protein epitopes, focusing on the envelope domain II fusion loop. Mutationally constrained epitope(s) on the surface of the yellow fever virus are identified. Appropriate antibody scaffolds that satisfy the epitope-paratope constraints are then identified, and the antibodies to target and neutralize the virus are engineered.

Anti-YFV antibodies are cloned, expressed and purified using known methods, and screened for expression and biophysical properties. The in silico designed antibodies are

software ZRANK) and rank-ordered based on shape complementarity (>0.6), buried surface area ($1,000 \text{ Sq. \AA}^2$) and potential to make contacts with the critical E-DII epitope residues (N106, K93 and K104 of E-protein; numbering correspond to primary sequence of 17D YF vaccine strain). Then, the CDR loops of the top ranking scaffolds were redesigned through Rosetta Antibody Design to make optimal contacts with the epitope. The engineered antibodies were screened for in vitro neutralization of YFV.

Synthetic DNA sequences encoding the VH and VL domains provided in Tables 1 and 2 were cloned in frame with an IgG1 constant region, into mammalian expression vector and various combinations of the heavy and light chains combined into at least one hundred and ten YFV antibodies. Examples of such combinations are shown in Table 3 below.

TABLE 3

YF antibody combinations										
	pVL.0	pVL.1	pVL.1.1	pVL.1.2	pVL.1.3	pVL.1.4	pVL.2	pVL.3	pVL.4	pVL.5
pVH.0	1	2					3	4		
pVH.1	5	6					7	8		
pVH.2	9	10					11	12		
pVH.3	13	14					15	16		
pVH.3.1			22	23	24	25				
pVH.3.2			26	27	28	29				
pVH.4									18	
pVH.5									19	
pVH.6										20
pVH.7										21

cloned in mammalian expression vectors and purified for further analysis. Both expression as well as various analytical parameters indicative of purity and stability are assessed. The potency of the antibodies against yellow fever virus is evaluated using in vitro and in vivo models of infection.

In vitro neutralization potency of engineered antibodies are assessed against the YF17D-204 vaccine strain of the yellow fever virus, using a plaque reduction neutralization test (PRNT). For selected antibodies, additional evaluation of potency are conducted using murine models of YFV infection.

Rational Design of Anti-YF mAbs

Promising anti-YFV monoclonal antibodies must neutralize a broad range of strains and target regions that are associated with robust protection. Epitopes play a critical role in the efficacy of a therapeutic antibody. In YFV infection, the envelope (E) protein is the predominant target of neutralizing antibodies. In order to define promising epitopes, a structure-guided analysis of the whole YFV assembly was conducted to identify spatially clustered, solvent accessible and sequence-conserved residues. This was done by mapping the sequence conservation scores (computed from an alignment of YFV envelope protein sequences) onto the homology model of a whole YFV E-protein assembly. Residues that have sequence conservation greater than 70% and solvent accessible surface area $>40\%$ were considered as putative epitope residues. From this analysis, the region proximal to the domain II (E-DII) hydrophobic fusion loop appeared highly accessible and conserved across YFV strains.

In order to engineer antibodies against the E-DII epitope, a structure-guided search for Fv scaffolds that recognize highly homologous epitope surfaces was conducted. Promising scaffolds were docked against the epitope (using the

Illustrative Examples of Expression Levels and Biophysical Properties

A subset of YFV antibodies were purified from larger scale transfections in Expi293 cells. Proteins from cell culture supernatants were purified using the HiTrap Protein A column. Following dialysis, the protein concentration was estimated by nanodrop using theoretical extinction coefficients. The yields of various antibodies are summarized in Table 4.

TABLE 4

Yield of YF mAbs expressed in Expi293 cells	
mAb	Expression mg/L
1	34.4
2	171
3	65.8

Additionally, purity and stability of various antibodies were determined by (1) UV spectrometry, (2) size exclusion chromatography, (3) microfluidic capillary electrophoresis and (4) thermal shift dye disassociation assay, respectively. Illustrative examples of such analytical biophysical parameters are summarized in Table 5 below.

TABLE 5

Purity and Stability of Select YF antibodies		
	1	2
Abs maxima 1 (nm)	228	228
Abs maxima 2 (nm)	278	279
E1% (g/100 mL)	14.2	13.8

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TABLE 5-continued

Purity and Stability of Select YF antibodies					1	2
HMW-1 (%)					0.16	0
HMW-2 (%)					0.68	0.62
Monomer (%)					99.04	98.99
LMW (%)					0.12	0.39
Reduced	LC	Size (kDa)	26.3	26.1		
		% of total	30.6	30.5		
	HC	Size (kDa)	58.5	58.4		
		% of total	68.7	69.5		
Intact	Size (kDa)		154.7	154.8		
	% of total		88.0	87.0		
Melting temperature T _m (° C.)					69.0	69.1

Additional characterization of the glycoforms of various antibodies were determined by procainamide-label assisted LC-FLD-MS. Illustrative examples of glycosylation profiles observed are below (Table 6).

TABLE 6

Glycosylation Profiles		
Glycan	1 (%)	2 (%)
Man5	0.8 ± 0.01	1.2 ± 0.07
G0	0.3 ± 0.00	0.5 ± 0.03
G0F	42.9 ± 0.05	51.6 ± 1.27
G1	0.1 ± 0.00	0.2 ± 0.02
G1F	29.4 ± 0.14	25.2 ± 0.12
G1'F	15.8 ± 0.04	15.3 ± 0.70
G2F	10.5 ± 0.12	6.0 ± 0.32
A1F	0.2 ± 0.00	0.1 ± 0.01

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Illustrative Examples of Neutralization Potency of Engineered Antibodies Against YF-17D

Plaque Reduction Neutralization Test (PRNT) was conducted to determine the neutralization efficacy of the YFV antibodies against the YFV (17D-204). Vero cells were infected with either virus alone, no virus or virus pre-incubated with various dilutions of YFV antibodies. Plates were incubated for 7 days, fixed and stained with crystal violet to visualize plaque formations. Neutralization curves were generated using the Prism software (FIG. 1) and the 50% effective concentration (EC50) values were calculated by nonlinear regression using a variable slope (Table 6). The neutralization efficacy of an antibody is directly proportional to plaque reduction and the potency is depicted as concentration at which 50% of the virus particles are neutralized. The antibodies were highly potent in neutralizing the virus and illustrative examples are shown below in Table 7.

TABLE 7

Summary of EC50 values for YF mAbs		
mAb	EC50 (µg/ml)	STDEV
1	0.012	0.006
2	0.007	0.002
3	0.011	0.010

Illustrative Examples of In Vivo Efficacy

FIG. 1 provides an illustrative example of in vivo efficacy of engineered antibodies against YF-17D-204 in a mouse model of infection. Efficacy of designed antibodies was tested in a lethal model of Yellow fever infection in AG129 mice. Protective efficacy of antibodies was tested in prophylaxis or as therapy as indicated in FIG. 1.

An illustrative example shown in FIG. 2, administration of antibody resulted in complete protection compared to control groups (FIG. 2A). Administration of antibody also led to greater than 2 Log₁₀ reduction in viremia at days 4 and 6 post infection (FIG. 2B).

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 73

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<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

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Glu Val Lys Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20 25 30

Ala Met Ser Trp Val Arg Gln Thr Pro Glu Lys Arg Leu Glu Trp Val
35 40 45

Ala Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys
50 55 60

Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Ile Leu Phe Leu
65 70 75 80

Gln Met Ser Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys Ala
85 90 95

Arg Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly

-continued

100	105	110
Gln Gly Thr Ser Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 2
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

 <400> SEQUENCE: 2

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg		
1	5	10
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asn Tyr		
20	25	30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Ser Leu Tyr Leu		
65	70	75
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys Ala		
85	90	95
Arg Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly		
100	105	110
Gln Gly Thr Thr Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 3
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

 <400> SEQUENCE: 3

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg		
1	5	10
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Thr Asp Tyr		
20	25	30
Tyr Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val		
35	40	45
Ser Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys		
50	55	60
Asp Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Ser Leu Tyr Leu		
65	70	75
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys Ala		
85	90	95
Arg Gly Asp Tyr Tyr Gly Thr Val Tyr Ser Ala Met Asp Tyr Trp Gly		
100	105	110
Gln Gly Thr Thr Val Thr Val Ser Ser		
115	120	

<210> SEQ ID NO 4
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 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 4

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Arg
1				5						10				15	

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Tyr	Ala	Phe	Thr	Asn	Tyr
			20					25					30		

Ala	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35						40					45			

Ser	Ser	Ile	Ser	Ser	Gly	His	Thr	Pro	Tyr	Tyr	Pro	Asp	Thr	Val	Lys
	50					55					60				

Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Lys	Ser	Leu	Tyr	Leu
65				70						75				80	

Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Leu	Tyr	Tyr	Cys	Ala
			85						90					95	

Arg	Gly	Asp	Tyr	Tyr	Gly	Ser	Ser	Tyr	Ser	Ala	Met	Asp	Tyr	Trp	Gly
		100						105					110		

Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser
	115						120	

<210> SEQ ID NO 5

<211> LENGTH: 120

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 5

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Arg
1				5						10				15	

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Tyr	Ala	Phe	Thr	Asn	Tyr
			20					25					30		

Gly	Val	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35					40						45			

Ser	Ser	Ile	Ser	Ser	Gly	Gly	Ser	Thr	Tyr	Tyr	Pro	Asp	Ser	Val	Lys
	50					55					60				

Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Lys	Ser	Leu	Tyr	Leu
65				70						75				80	

Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Leu	Tyr	Tyr	Cys	Ala
			85						90					95	

Arg	His	Asp	Tyr	Tyr	Gly	Ser	Ser	Tyr	Ser	Ala	Met	Asp	Tyr	Trp	Gly
		100						105					110		

Gly	Thr	Thr	Val	Thr	Val	Ser	Ser
	115					120	

<210> SEQ ID NO 6

<211> LENGTH: 121

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 6

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Arg
1				5						10				15	

Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Tyr	Ala	Phe	Thr	Asn	Tyr
			20					25					30		

Gly	Val	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
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35	40	45
Ser Ser Ile Ser Ser Gly Gly Ser Thr Tyr Tyr Pro Asp Ser Val Lys		
50	55	60
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Ser Leu Tyr Leu		
65	70	75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys Ala		
	85	90 95
Arg Ser His Tyr Tyr Gly Ser Ser Tyr Asp Ala Met Asp Tyr Trp Gly		
	100	105 110
Gln Gly Thr Thr Val Thr Val Ser Ser		
	115	120

<210> SEQ ID NO 7
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 7

Glu Val Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Pro Glu Trp Met
35 40 45
Gly Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys
50 55 60
Gly Gly Arg Val Thr Ile Thr Ala Asp Asp Phe Ala Gly Thr Val Tyr
65 70 75 80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys
85 90 95
Arg Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly
100 105 110
Lys Gly Thr Thr Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 8
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 8

Glu Val Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Thr Phe Thr Asn Tyr
20 25 30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Pro Glu Trp Met
35 40 45
Gly Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys
50 55 60
Gly Arg Val Thr Ile Thr Ala Asp Asp Phe Ala Gly Thr Val Tyr Met
65 70 75 80
Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys Ala
85 90 95

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Arg Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly
 100 105 110

Lys Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 9
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 9

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Gly Phe Thr Phe Thr Asn Tyr Ala
 20 25 30
 Met Ser Trp Val Arg Gln Ala Pro Glu Gln Gly Leu Glu Trp Met Gly
 35 40 45
 Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys Gly
 50 55 60
 Arg Val Thr Met Thr Ala Asp Thr Ser Thr Asn Thr Ala Tyr Met Glu
 65 70 75 80
 Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys Ala Arg
 85 90 95
 Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 10
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 10

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ser Ala Ser Gly Phe Thr Phe Thr Asn Tyr
 20 25 30
 Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Tyr Val
 35 40 45
 Ser Ser Ile Ser Ser Gly His Thr Pro Tyr Tyr Pro Asp Ser Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr Phe
 65 70 75 80
 Glu Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys Val
 85 90 95
 Arg Gly Asp Tyr Tyr Gly Ser Val Tyr Ser Ala Met Asp Tyr Trp Gly
 100 105 110
 Gln Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 11
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 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 11

Ala Ser Val Leu Ser Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu
1          5          10          15
Val Lys Pro Gly Ala Ser Val Lys Leu Ser Cys Lys Thr Ser Glu Asn
20          25          30
Thr Phe Thr Glu Tyr Thr Met His Trp Val Lys Gln Ser His Gly Lys
35          40          45
Ser Leu Glu Trp Ile Gly Gly Ile Asp Pro Asn Asn Gly Gly Thr Asn
50          55          60
Tyr Asn Gln Lys Phe Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser
65          70          75          80
Ser Asn Thr Ala Tyr Met Glu Leu Arg Ser Leu Thr Ser Glu Asp Ser
85          90          95
Ala Val Tyr Tyr Cys Gly Arg Arg Asp Tyr Tyr Ala Leu Asp Tyr Trp
100         105         110
Gly Gln Gly Thr Ser Val Thr Val Ala Ser
115          120

<210> SEQ ID NO 12
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 12

Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
20          25          30
Trp Met His Trp Val Lys Leu Arg Pro Gly Gln Gly Phe Glu Trp Ile
35          40          45
Gly Asp Ile Asn Pro Asn Asn Gly Gly Pro Ser Tyr Asn Glu Lys Phe
50          55          60
Lys Arg Lys Ala Thr Leu Thr Val Asp Thr Ser Ser Ser Thr Ala Tyr
65          70          75          80
Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85          90          95
Thr Ile Asp Asp Gly Tyr Arg Phe Gly Tyr Trp Gly Gln Gly Thr Leu
100         105         110
Val Thr Val Ser Ala
115

<210> SEQ ID NO 13
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 13

Gln Val Gln Leu Gln Gln Ser Gly Ser Glu Leu Met Lys Pro Gly Ala
1          5          10          15
Ser Val Gln Ile Ser Cys Lys Ala Thr Gly Tyr Thr Phe Ser Asp Tyr
20          25          30

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-continued

Trp Ile Glu Trp Val Lys Gln Arg Pro Gly His Gly Leu Glu Trp Ile
 35 40 45

Gly Asp Ile Leu Cys Gly Thr Gly Arg Thr Arg Tyr Asn Glu Lys Leu
 50 55 60

Lys Ala Met Ala Thr Phe Thr Ala Asp Thr Ser Ser Asn Thr Ala Phe
 65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ser Ala Ser Tyr Gly Asp Tyr Ala Asp Tyr Trp Gly His Gly
 100 105 110

Thr Thr Leu Thr Val Ser Ser
 115

<210> SEQ ID NO 14
 <211> LENGTH: 115
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 14

Ala Gln Leu Gln Gln Ser Gly Thr Gly Leu Ala Arg Pro Gly Ala Ser
 1 5 10 15

Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr Gly
 20 25 30

Ile Ser Trp Val Thr Gln Arg Ala Gly Gln Gly Leu Glu Trp Ile Gly
 35 40 45

Val Ile Tyr Pro Arg Ser Gly Asn Thr Tyr Tyr Asn Glu Lys Phe Arg
 50 55 60

Gly Lys Ala Thr Leu Thr Ala Asp Lys Ser Ser Ser Ser Ala Tyr Met
 65 70 75 80

Glu Leu Arg Gly Leu Thr Ala Glu Asp Ser Ala Val Tyr Phe Cys Ala
 85 90 95

Arg Glu Asn Tyr Gly Ser Val Tyr Trp Gly Gln Gly Thr Thr Leu Thr
 100 105 110

Val Ser Ser
 115

<210> SEQ ID NO 15
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 15

Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Thr
 20 25 30

Tyr Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Arg Ile Asp Pro Ala Asn Gly Tyr Ser Lys Tyr Asp Pro Lys Phe
 50 55 60

Gln Gly Lys Ala Thr Ile Thr Ala Asp Thr Ser Ser Asn Ala Ala Tyr
 65 70 75 80

Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95

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Ala Arg Asp Tyr Glu Gly Phe Ala Tyr Trp Gly Gln Gly Thr Leu Val
 100 105 110

Thr Val Ser Ser
 115

<210> SEQ ID NO 16
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 16

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Tyr
 20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Val Ile Trp Tyr Asp Gly Ser Lys Thr Tyr Tyr Gly Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Lys Asp Asn Ser Lys Lys Met Val Asn
 65 70 75 80

Leu Gln Met Asp Ser Leu Gly Val Glu Asp Thr Ala Phe Tyr Tyr Cys
 85 90 95

Ala Arg Gly Ile Ala Gly Gly Trp Ala Phe Trp Gly Ile Asp Leu Trp
 100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 17
 <211> LENGTH: 116
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 17

Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Thr
 20 25 30

Tyr Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Arg Ile Asp Pro Ala Asn Gly Tyr Ser Lys Tyr Asp Pro Lys Phe
 50 55 60

Gln Gly Lys Ala Thr Ile Thr Ala Asp Thr Ser Ser Asn Ala Ala Tyr
 65 70 75 80

Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95

Ala Arg Asp Tyr Glu Gly Phe Ala Tyr Trp Gly Gln Gly Thr Leu Val
 100 105 110

Thr Val Ser Ser
 115

<210> SEQ ID NO 18
 <211> LENGTH: 123
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 18

Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Arg Lys Pro Gly Ala
1             5             10            15

Ser Thr Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr His Tyr
20            25            30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
35            40            45

Gly Ile Ile Asn Pro Ser Gly Gly Ser Thr Thr Tyr Ala Gln Lys Leu
50            55            60

Gln Gly Arg Val Thr Met Thr Arg Asp Thr Ser Thr Ser Thr Val Tyr
65            70            75            80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85            90            95

Ala Arg Asp Trp Gly Ser Asn Tyr Val Trp Gly Ser Tyr Pro Lys Tyr
100           105           110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115            120

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<210> SEQ ID NO 19
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 19

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg
1             5             10            15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Asp Tyr
20            25            30

Ala Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35            40            45

Ser Thr Ile Ser Trp Asn Ser Gly Ser Ile Gly Tyr Ala Asp Ser Val
50            55            60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Lys Ser Leu Tyr
65            70            75            80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys
85            90            95

Ala Lys Asp Ile Gln Tyr Gly Asn Tyr Tyr Tyr Gly Met Asp Val Trp
100           105           110

Gly Gln Gly Thr Thr Val Thr Val Ser Ser
115            120

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<210> SEQ ID NO 20
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 20

Asp Val Gln Leu Val Glu Pro Gly Ala Glu Leu Val Gln Pro Gly Ala
1             5             10            15

Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Ser Ser Tyr
20            25            30

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-continued

Trp Ile Asn Trp Glu Lys Gln Arg Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45

Gly Asn Ile Tyr Pro Gly Ser Gly Thr Val Asn Tyr Asp Asp Lys Phe
 50 55 60

Lys Ser Lys Ala Thr Leu Thr Ile Asp Thr Ser Ser Asn Thr Ala Tyr
 65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Thr Arg Gly Gly Ser His Ala Met Asp Tyr Trp Gly Gln Gly Thr Ser
 100 105 110

Val Thr Val Ser Ser
 115

<210> SEQ ID NO 21
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 21

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Tyr Ser Asn His
 20 25 30

Trp Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Val Trp Val
 35 40 45

Ser Arg Ile Asn Ser Asp Gly Ser Thr Arg Asn Tyr Ala Asp Phe Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Glu Asn Thr Leu Tyr
 65 70 75 80

Leu Glu Met Asn Ser Leu Thr Ala Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Arg Asp Gly Val Arg Phe Tyr Tyr Asp Ser Thr Gly Tyr Tyr Pro
 100 105 110

Asp Ser Phe Phe Lys Tyr Gly Met Asp Val Trp Gly Gln Gly Thr Thr
 115 120 125

Val Thr Val Ser Ser
 130

<210> SEQ ID NO 22
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 22

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Tyr Ser Asn His
 20 25 30

Trp Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Val Trp Val
 35 40 45

Ser Arg Ile Asn Ser Asp Gly Ser Thr Arg Asn Tyr Ala Asp Phe Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Glu Asn Thr Leu Tyr

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65	70	75	80
Leu Glu Met Asn Ser	Leu Thr Ala Asp	Asp Thr Ala Val Tyr Tyr Cys	
	85	90	95
Val Arg Asp Gly Val Arg Phe Tyr	Tyr Asp Ser Thr Gly Tyr Tyr Pro		
	100	105	110
Asp Ser Phe Phe Lys Tyr Gly Met Asp Val Trp Gly Gln Gly Thr Thr			
	115	120	125
Val Thr Val Ser Ser			
	130		

<210> SEQ ID NO 23
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 23

Glu Val Lys Leu Val Glu Ser Gly Gly Gly Leu Val Leu Pro Gly Gly	
1	5 10 15
Ser Leu Arg Leu Ser Cys Ala Thr Ser Gly Phe Thr Phe Thr Asp Tyr	
	20 25 30
Tyr Met Thr Trp Val Arg Gln Pro Pro Gly Lys Ala Leu Glu Trp Leu	
	35 40 45
Gly Phe Ile Gly Asn Lys Ala Asn Asp Tyr Thr Thr Glu Tyr Ser Ala	
	50 55 60
Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Gln Ser Ile	
65	70 75 80
Leu Tyr Leu Gln Met Ser Thr Leu Arg Ala Glu Asp Arg Ala Thr Tyr	
	85 90 95
Tyr Cys Ala Thr Val Tyr Gly Asn Tyr Pro Tyr Phe Asp Val Trp Gly	
	100 105 110
Ala Gly Thr Thr Val Ala Val Ser Ser	
	115 120

<210> SEQ ID NO 24
 <211> LENGTH: 128
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 24

Glu Val Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ser	
1	5 10 15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Thr Phe Asn Asn Tyr	
	20 25 30
Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met	
	35 40 45
Gly Gly Ile Ile Pro Ile Phe Gly Gly Ala Asn Tyr Ala Gln Lys Phe	
	50 55 60
Gln Gly Arg Val Thr Ile Thr Ala Asp Arg Ser Thr Ser Thr Val Tyr	
65	70 75 80
Met Glu Leu Ser Gly Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys	
	85 90 95
Ala Arg Arg Pro Gln Ser Ile Phe Asp Trp Asn Phe Asp Leu Trp Gly	
	100 105 110

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Arg Gly Thr Leu Val Thr Val Ser Ser Ala Gly Thr Lys Gly Pro Ser
 115 120 125

<210> SEQ ID NO 25
 <211> LENGTH: 134
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 25

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser His
 20 25 30
 Trp Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Val Trp Val
 35 40 45
 Ser Arg Thr Asn Ser Asp Gly Ser Ser Thr Ser Tyr Ala Asp Ser Val
 50 55 60
 Lys Gly Arg Phe Met Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr
 65 70 75 80
 Leu His Met Asn Gly Leu Arg Ala Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95
 Ala Arg Asp Gly Val Arg Tyr Tyr Tyr Asp Ser Thr Gly Tyr Tyr Pro
 100 105 110
 Asp Asn Phe Phe Gln Tyr Gly Leu Asp Val Trp Gly Gln Gly Thr Thr
 115 120 125
 Val Thr Val Ser Ser Ala
 130

<210> SEQ ID NO 26
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 26

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asp Tyr
 20 25 30
 Asn Met His Trp Val Lys Gln Ser His Gly Lys Ser Leu Glu Trp Val
 35 40 45
 Gly Tyr Thr Tyr Pro Tyr Asn Gly Gly Ile Gly Tyr Asn Gln Lys Phe
 50 55 60
 Lys Ser Lys Ala Thr Leu Thr Leu Asp Asn Ser Ser Arg Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Arg Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95
 Val Arg Arg Gly Tyr Arg Tyr Asp Gly Ala His Phe Asp Tyr Trp Gly
 100 105 110
 Gln Gly Thr Thr Leu Thr Val Ser Ser
 115 120

<210> SEQ ID NO 27
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 27

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Arg Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
          20          25          30
Gly Met His Trp Val Arg Gln Ala Pro Glu Lys Gly Leu Glu Trp Val
          35          40          45
Ala Tyr Ile Ser Ser Gly Ser Ser Thr Leu His Tyr Ala Asp Thr Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Pro Lys Asn Thr Leu Phe
          65          70          75          80
Leu Gln Met Thr Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys
          85          90          95
Ala Arg Trp Gly Asn Tyr Pro His Tyr Ala Met Asp Tyr Trp Gly Gln
          100          105          110
Gly Thr Ser Val Thr Val Ser Ser
          115          120

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<210> SEQ ID NO 28
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 28

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Val Ser Cys Lys Ala Gly Phe Asn Ile Lys Asp Val Tyr
          20          25          30
Met Ser Trp Val Arg Gln Ala Pro Glu Gln Gly Leu Glu Trp Met Gly
          35          40          45
Arg Ile Asp Pro Glu Asn Gly Asp Thr Lys Tyr Asp Pro Lys Leu Gln
          50          55          60
Gly Arg Val Thr Met Thr Ala Asp Thr Ser Thr Asn Thr Ala Tyr Met
          65          70          75          80
Glu Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys Ala
          85          90          95
Arg Gly Trp Glu Gly Phe Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr
          100          105          110
Val Ser Ser
          115

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<210> SEQ ID NO 29
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 29

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ser Ala Ser Gly Phe Thr Phe Ser Thr Tyr
          20          25          30

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Ser Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Tyr Val
 35 40 45

Ser Ala Ile Thr Gly Glu Gly Asp Ser Ala Phe Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80

Phe Glu Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Gly Gly Tyr Ser Asn Phe Tyr Tyr Tyr Thr Met Asp Val Trp
 100 105 110

Gly Gln Gly Thr Thr Val Thr Val Ser Ser Gly
 115 120

<210> SEQ ID NO 30
 <211> LENGTH: 133
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 30

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Arg Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Tyr Ser Asn His
 20 25 30

Trp Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Val Trp Val
 35 40 45

Ser Arg Ile Asn Ser Asp Gly Ser Thr Arg Asn Tyr Ala Asp Phe Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Glu Asn Thr Leu Tyr
 65 70 75 80

Leu Glu Met Asn Ser Leu Thr Ala Asp Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Val Arg Asp Gly Val Arg Phe Tyr Tyr Asp Ser Thr Gly Tyr Tyr Pro
 100 105 110

Asp Ser Phe Phe Lys Tyr Gly Met Asp Val Trp Gly Gln Gly Thr Thr
 115 120 125

Val Thr Val Ser Ser
 130

<210> SEQ ID NO 31
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 31

Asp Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Arg Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Phe
 20 25 30

Gly Met His Trp Val Arg Gln Ala Pro Glu Lys Gly Leu Glu Trp Val
 35 40 45

Ala Tyr Ile Ser Ser Gly Ser Ser Thr Leu His Tyr Ala Asp Thr Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Pro Lys Asn Thr Leu Phe
 65 70 75 80

-continued

Leu Gln Met Thr Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys
 85 90 95

Ala Arg Trp Gly Asn Tyr Pro His Tyr Ala Met Asp Tyr Trp Gly Gln
 100 105 110

Gly Thr Ser Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 32
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 32

Gln Val Gln Leu Gln Gln Pro Gly Ala Glu Leu Val Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr
 20 25 30

Asn Met His Trp Val Lys Gln Thr Pro Gly Arg Gly Leu Glu Trp Ile
 35 40 45

Gly Ala Ile Tyr Pro Gly Asn Gly Asp Thr Ser Tyr Asn Gln Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Ala Asp Lys Ser Ser Ser Thr Ala Tyr
 65 70 75 80

Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Ser Thr Tyr Tyr Gly Gly Asp Trp Tyr Phe Asn Val Trp Gly
 100 105 110

Ala Gly Thr Thr Val Thr Val Ser Ala
 115 120

<210> SEQ ID NO 33
 <211> LENGTH: 119
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 33

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ala Phe Ser Tyr Ser
 20 25 30

Trp Ile Asn Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Arg Ile Phe Pro Gly Asp Gly Asp Thr Asp Tyr Asn Gly Lys Phe
 50 55 60

Lys Gly Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ala Arg Asn Val Phe Asp Gly Tyr Trp Leu Val Tyr Trp Gly Gln Gly
 100 105 110

Thr Leu Val Thr Val Ser Ser
 115

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<210> SEQ ID NO 34
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 34

Glu Val Gln Leu Val Glu Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Gly Pro Phe Arg Ser Tyr
 20 25 30
 Ala Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Pro Glu Trp Met
 35 40 45
 Gly Gly Ile Ile Pro Ile Phe Gly Thr Thr Lys Tyr Ala Pro Lys Phe
 50 55 60
 Gln Gly Arg Val Thr Ile Thr Ala Asp Asp Phe Ala Gly Thr Val Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Met Tyr Tyr Cys
 85 90 95
 Ala Lys His Met Gly Tyr Gln Val Arg Glu Thr Met Asp Val Trp Gly
 100 105 110
 Lys Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 35
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 35

Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr
 20 25 30
 Ile Met Met Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Ser Ile Tyr Pro Ser Gly Gly Ile Thr Phe Tyr Ala Asp Thr Val
 50 55 60
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Arg Ile Lys Leu Gly Thr Val Thr Thr Val Asp Tyr Trp Gly Gln
 100 105 110
 Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 36
 <211> LENGTH: 120
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 36

Gln Val Thr Leu Arg Glu Ser Gly Pro Ala Leu Val Lys Pro Thr Gln
 1 5 10 15

-continued

Thr Leu Thr Leu Thr Cys Thr Phe Ser Gly Phe Ser Leu Ser Thr Ala
 20 25 30
 Gly Met Ser Val Gly Trp Ile Arg Gln Pro Pro Gly Lys Ala Leu Glu
 35 40 45
 Trp Leu Ala Asp Ile Trp Trp Asp Asp Lys Lys His Tyr Asn Pro Ser
 50 55 60
 Leu Lys Asp Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys Asn Gln Val
 65 70 75 80
 Val Leu Lys Val Thr Asn Met Asp Pro Ala Asp Thr Ala Thr Tyr Tyr
 85 90 95
 Cys Ala Arg Asp Met Ile Phe Asn Phe Tyr Phe Asp Val Trp Gly Gln
 100 105 110
 Gly Thr Thr Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 37
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 37

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Met Tyr Ala Ser Leu Gly
 1 5 10 15
 Glu Arg Val Thr Val Thr Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
 20 25 30
 Leu Ser Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
 35 40 45
 Tyr Arg Ala Asn Arg Leu Phe Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Gln Asp Tyr Ser Leu Thr Ile Ser Ser Leu Glu Tyr
 65 70 75 80
 Glu Asp Met Gly Ile Phe Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Phe
 85 90 95
 Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 38
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 38

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
 20 25 30
 Leu Ser Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
 35 40 45
 Tyr Arg Ala Asn Arg Leu Phe Asp Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Ile Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Phe

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85	90	95
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Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 39
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 39

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
20 25 30
Leu Asn Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
35 40 45
Tyr Arg Val Asn Arg Leu Val Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Ile Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Tyr
85 90 95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 40
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 40

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Lys Ser Tyr
20 25 30
Leu Ser Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
35 40 45
Tyr Arg Val Asn Arg Leu Val Asp Gly Val Pro Ser Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Ala Ile Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Tyr
85 90 95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 41
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 41

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

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Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
      20                      25                      30
Leu Asn Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
      35                      40                      45
Tyr Arg Val Asn Arg Leu Val Asp Gly Val Pro Ser Arg Phe Ser Gly
      50                      55                      60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
      65                      70                      75                      80
Glu Asp Phe Ala Ile Tyr Tyr Cys Leu His Tyr Asp Glu Phe Pro Tyr
      85                      90                      95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
      100                      105

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<210> SEQ ID NO 42
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 42

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Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1       5              10              15
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Lys Ser Tyr
      20                      25                      30
Leu Ser Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
      35                      40                      45
Tyr Arg Val Asn Arg Leu Val Asp Gly Val Pro Ser Arg Phe Ser Gly
      50                      55                      60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
      65                      70                      75                      80
Glu Asp Phe Ala Ile Tyr Tyr Cys Leu His Tyr Asp Glu Phe Pro Tyr
      85                      90                      95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
      100                      105

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<210> SEQ ID NO 43
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 43

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Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1       5              10              15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Ile Asn Ser Tyr
      20                      25                      30
Leu Ser Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
      35                      40                      45
Tyr Arg Ala Asn Arg Leu Met Ile Gly Val Pro Ser Arg Phe Ser Gly
      50                      55                      60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
      65                      70                      75                      80
Glu Asp Phe Ala Ile Tyr Tyr Cys Leu Gln Tyr Asp Asp Phe Pro Leu
      85                      90                      95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys

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100              105

<210> SEQ ID NO 44
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 44

Asp Ile Arg Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1              5              10              15

Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asp Ile Asn Ser Phe
20              25              30

Leu Thr Trp Leu Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
35              40              45

Tyr Arg Ala Asn Arg Val Phe Asp Gly Val Pro Ser Arg Phe Ser Gly
50              55              60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65              70              75              80

Glu Asp Phe Ala Ile Tyr Tyr Cys Leu Gln Tyr Asp Asp Phe Pro Leu
85              90              95

Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys
100              105

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<210> SEQ ID NO 45
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 45

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
1              5              10              15

Lys Val Thr Ile Ser Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr Leu
20              25              30

Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu Ile Tyr
35              40              45

Arg Ala Asn Arg Leu Phe Asp Gly Ile Pro Asp Arg Phe Ser Gly Ser
50              55              60

Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile Thr Gly Leu Gln Thr Gly
65              70              75              80

Asp Glu Ala Asn Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Phe Thr
85              90              95

Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100              105

```

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<210> SEQ ID NO 46
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 46

Asp Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1              5              10              15

Gln Arg Ala Thr Ile Ser Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
20              25              30

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Leu Ser Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
   35                               40               45

Tyr Arg Ala Asn Arg Leu Phe Asp Gly Val Pro Asp Arg Phe Ser Gly
   50                               55               60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Ala
   65                               70               75               80

Glu Asp Val Ala Val Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Phe
                        85               90               95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg
      100               105

```

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<210> SEQ ID NO 47
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 47

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Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro Gly
 1           5           10           15

Glu Arg Ala Thr Leu Ser Cys Lys Ala Ser Gln Asp Ile Asn Ser Tyr
 20           25           30

Leu Ser Trp Tyr Gln His Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile
 35           40           45

Tyr Arg Ala Asn Arg Leu Phe Asp Gly Val Pro Ala Arg Phe Ser Gly
 50           55           60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Thr Leu Glu Pro
 65           70           75           80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Leu Gln Tyr Asp Glu Phe Pro
 85           90           95

Phe Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100           105

```

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<210> SEQ ID NO 48
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 48

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Asp Ile Val Met Thr Gln Ser Gln Lys Phe Met Ser Thr Ser Val Gly
 1           5           10           15

Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln His Val Gly Ser Ala
 20           25           30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Thr Leu Leu Ile
 35           40           45

His Ser Ala Ser Asn Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Gly
 50           55           60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Ile Gln Ser
 65           70           75           80

Glu Asp Leu Ala Asp Tyr Phe Cys Gln Gln Tyr Asn Ser Tyr Pro Thr
 85           90           95

Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100           105

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<210> SEQ ID NO 49
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 49

Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Leu Ser Ala Ser Val Gly
1 5 10 15
Glu Thr Val Thr Ile Thr Cys Arg Ala Ser Gly Asn Ile His Asn Tyr
20 25 30
Leu Ala Trp Tyr Gln Gln Lys Gln Gly Lys Ser Pro Gln Leu Leu Val
35 40 45
Tyr Asn Ala Lys Thr Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Ala
50 55 60
Ser Gly Ser Gly Thr Gln Tyr Ser Leu Lys Ile Asn Ser Leu Gln Pro
65 70 75 80
Glu Asp Phe Gly Ser Tyr Tyr Cys Gln His Phe Trp Ser Thr Pro Arg
85 90 95
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg
100 105

<210> SEQ ID NO 50
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 50

Asp Ile Val Met Thr Gln Ser His Lys Phe Met Ser Thr Ser Val Gly
1 5 10 15
Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln Asp Val Ser Thr Ala
20 25 30
Val Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Lys Leu Leu Ile
35 40 45
Ser Trp Ala Ser Thr Arg His Thr Gly Val Pro Asp Arg Phe Thr Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Val Gln Ala
65 70 75 80
Glu Asp Leu Ala Leu Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Leu
85 90 95
Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
100 105

<210> SEQ ID NO 51
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 51

Asp Ile Val Met Thr Gln Ser Gln Lys Phe Met Ser Thr Ser Val Gly
1 5 10 15
Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln Asn Val Gly Thr Ala
20 25 30
Val Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Lys Leu Leu Ile
35 40 45

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Tyr Ser Ala Ser Asn Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Met Gln Ser
 65 70 75 80
 Glu Asp Leu Ala Asp Tyr Phe Cys Gln Gln Phe Ser Ser Tyr Pro Tyr
 85 90 95
 Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 52
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 52

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
 1 5 10 15
 Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Glu Ser Val Val Arg Tyr
 20 25 30
 Gly Asn Ser Phe Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
 35 40 45
 Lys Leu Leu Ile Tyr Arg Ala Ser Ser Leu Glu Ser Gly Ile Pro Thr
 50 55 60
 Arg Phe Ser Gly Ser Gly Ser Arg Thr Asp Phe Thr Leu Thr Ile Asn
 65 70 75 80
 Pro Val Glu Ala Asp Asp Val Ala Thr Tyr Tyr Cys Gln Gln Thr Asn
 85 90 95
 Val Asp Pro Trp Ala Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 53
 <211> LENGTH: 108
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 53

Asp Val Val Met Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
 1 5 10 15
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Val Tyr Ser Tyr
 20 25 30
 Leu Gly Trp Tyr Gln His Lys Pro Gly Arg Ser Pro Arg Leu Leu Ile
 35 40 45
 Phe Gly Val Thr Ser Arg Ala Thr Gly Val Pro Asp Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro
 65 70 75 80
 Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Ala Gly Ser Ala Tyr
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
 100 105

<210> SEQ ID NO 54
 <211> LENGTH: 111
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 54
Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1           5           10           15
Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Glu Ser Val Val Arg Tyr
20           25           30
Gly Asn Ser Phe Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
35           40           45
Lys Leu Leu Ile Tyr Arg Ala Ser Ser Leu Glu Ser Gly Ile Pro Thr
50           55           60
Arg Phe Ser Gly Ser Gly Ser Arg Thr Asp Phe Thr Leu Thr Ile Asn
65           70           75           80
Pro Val Glu Ala Asp Asp Val Ala Thr Tyr Tyr Cys Gln Gln Thr Asn
85           90           95
Val Asp Pro Trp Ala Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100          105          110

<210> SEQ ID NO 55
<211> LENGTH: 111
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 55
Gln Ser Val Leu Thr Gln Pro Ser Ser Val Ser Gly Thr Pro Gly Gln
1           5           10           15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20           25           30
Thr Val Asn Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
35           40           45
Ile Tyr Gly Asn Asn Gln Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50           55           60
Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Gln
65           70           75           80
Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp Asp Ser Leu
85           90           95
Asn Gly Pro Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly
100          105          110

<210> SEQ ID NO 56
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 56
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Arg Thr Ser Gln Ser Ile Ser Ser Tyr
20           25           30
Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35           40           45
Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50           55           60

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Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Arg
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 57
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 57

Asp Ile Val Met Thr Gln Ser Gln Lys Phe Met Ser Thr Ser Val Gly
1 5 10 15

Asp Arg Val Ser Ile Thr Cys Lys Ala Ser Gln Asn Val Arg Thr Ser
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Lys Ala Leu Ile
35 40 45

Tyr Leu Ala Ser Asn Arg His Thr Gly Val Pro Asp Arg Phe Thr Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Val Gln Ser
65 70 75 80

Glu Asp Leu Ala Asp Tyr Phe Cys Leu Gln His Trp Thr Tyr Pro Tyr
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 58
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 58

Gly Ala Ser Gln Ser Val Leu Thr Gln Pro Val Ser Val Ser Gly Ser
1 5 10 15

Pro Gly Gln Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asn Ala
20 25 30

Asp Thr Tyr Asn Leu Val Ser Trp Tyr Gln Gln Arg Pro Gly Lys Ala
35 40 45

Pro Lys Leu Met Ile Tyr Glu Gly Thr Lys Arg Pro Ser Gly Val Ser
50 55 60

Asn Arg Phe Ser Ala Ser Lys Ser Ala Thr Ala Ala Ser Leu Thr Ile
65 70 75 80

Ser Gly Leu Gln Pro Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr
85 90 95

Ala Thr Ser Arg Thr Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val
100 105 110

Val

<210> SEQ ID NO 59
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 59

Gly Ala Ser Gln Ser Val Leu Thr Gln Pro Val Ser Val Ser Gly Ser
1          5              10              15
Pro Gly Gln Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asn Ala
          20              25              30
Asp Thr Tyr Asn Leu Val Ser Trp Tyr Gln Gln Arg Pro Gly Lys Ala
          35              40              45
Pro Lys Leu Met Ile Tyr Glu Gly Thr Lys Arg Pro Ser Gly Val Ser
          50              55              60
Asn Arg Phe Ser Ala Ser Lys Ser Ala Thr Ala Ala Ser Leu Thr Ile
65          70              75              80
Ser Gly Leu Gln Pro Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr
          85              90              95
Ala Thr Ser Arg Thr Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val
          100              105              110
Val Ala Ala
          115

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<210> SEQ ID NO 60
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 60

Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1          5              10              15
Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
          20              25              30
His Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Ile Trp Ile Tyr
          35              40              45
Glu Ser Ser Lys Leu Ala Ser Gly Val Pro Val Arg Phe Ser Gly Ser
          50              55              60
Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
65          70              75              80
Asp Val Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser His Pro His Pro
          85              90              95
Leu Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
          100              105

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<210> SEQ ID NO 61
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 61

Gln Ser Val Leu Thr Gln Pro Pro Ser Ala Ser Gly Thr Pro Gly Gln
1          5              10              15
Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Val Gly Ser Asn
          20              25              30
Tyr Val Tyr Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Leu Leu
          35              40              45

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Ile Tyr Arg Asn Asn Arg Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
 50                      55                      60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Ser Gly Leu Arg
 65                      70                      75                      80

Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Thr Trp Asp Asp Ser Leu
                      85                      90                      95

Ser Gly Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln
 100                      105                      110

Pro Lys Ala
 115

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<210> SEQ ID NO 62
<211> LENGTH: 115
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 62

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Arg Ser Gln Ser Ala Leu Thr Gln Pro Ala Ser Val Ser Gly Ser Pro
 1                      5                      10                      15

Gly Gln Ser Ile Thr Ile Ser Cys Thr Gly Ile Ser Ser Asp Val Glu
 20                      25                      30

Thr Tyr Asn Leu Val Ser Trp Tyr Glu Gln His Pro Gly Lys Ala Pro
 35                      40                      45

Lys Leu Ile Ile Tyr Glu Ala Ser Lys Arg Pro Ser Gly Val Ser Asn
 50                      55                      60

Arg Phe Ser Gly Ser Lys Ser Gly Asn Thr Ala Ser Leu Ala Ile Ser
 65                      70                      75                      80

Gly Leu Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr Ala
 85                      90                      95

Gly Gly Lys Ser Leu Val Phe Gly Gly Gly Thr Arg Leu Thr Val Leu
 100                      105                      110

Gly Gln Pro
 115

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<210> SEQ ID NO 63
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide

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<400> SEQUENCE: 63

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Asp Ile Lys Met Thr Gln Ser Pro Ser Ser Met Tyr Ala Ser Leu Gly
 1                      5                      10                      15

Glu Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Gly Ile Asn Ser Asp
 20                      25                      30

Leu Ser Trp Phe Gln Gln Lys Pro Gly Lys Ser Pro Lys Thr Leu Ile
 35                      40                      45

Tyr Arg Ala Asn Arg Leu Val Asp Gly Val Pro Ser Arg Phe Ser Gly
 50                      55                      60

Ser Gly Ser Gly Gln Asp Tyr Ser Leu Thr Ile Ser Ser Leu Glu Tyr
 65                      70                      75                      80

Glu Asp Met Gly Ile Tyr Tyr Cys Leu Gln Tyr Asp Glu Phe Pro Leu
 85                      90                      95

Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
 100                      105

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<210> SEQ ID NO 64
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 64

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Asn Ile Val Met Thr Gln Ser Pro Lys Ser Met Ser Met Ser Val Gly
1           5           10           15
Glu Arg Val Thr Leu Thr Cys Lys Ala Ser Glu Asn Val Gly Thr Tyr
20          25          30
Val Ser Trp Tyr Gln Gln Lys Pro Glu Gln Ser Pro Lys Leu Leu Ile
35          40          45
Tyr Gly Ala Ser Asn Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Gly
50          55          60
Ser Gly Ser Ala Thr Asp Phe Thr Leu Thr Ile Ser Ser Val Gln Ala
65          70          75          80
Glu Asp Leu Ala Asp Tyr His Cys Gly Gln Ser Tyr Ser Thr Pro Tyr
85          90          95
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100         105

```

<210> SEQ ID NO 65
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 65

```

Asp Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1           5           10           15
Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Glu Asn Val Asp Lys Tyr
20          25          30
Gly Asn Ser Phe Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
35          40          45
Lys Leu Leu Ile Tyr Arg Ala Ser Glu Leu Gln Trp Gly Val Pro Asp
50          55          60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
65          70          75          80
Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Arg Ser Asn
85          90          95
Glu Val Pro Trp Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg
100         105         110
Thr Val Ala His His His His His His
115         120

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<210> SEQ ID NO 66
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 66

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Ser Glu Ile Val Leu Thr Gln Ser Pro Ala Thr Leu Ser Leu Ser Pro
1           5           10           15

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-continued

Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Ile Ser Thr
 20 25 30
 Phe Leu Ala Trp Tyr Gln His Lys Pro Gly Gln Ala Pro Arg Leu Leu
 35 40 45
 Ile Tyr Asp Ala Ser Thr Arg Ala Thr Gly Val Pro Ala Arg Phe Ser
 50 55 60
 Gly Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Thr Leu Glu
 65 70 75 80
 Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Arg Tyr Asn Trp Pro
 85 90 95
 Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105

<210> SEQ ID NO 67
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 67

Gln Ser Val Leu Thr Gln Pro Val Ser Val Ser Gly Ser Pro Gly Gln
 1 5 10 15
 Ser Ile Thr Ile Ser Cys Thr Gly Thr Ser Ser Asn Ala Asp Thr Tyr
 20 25 30
 Asn Leu Val Ser Trp Tyr Gln Gln Arg Pro Gly Lys Ala Pro Lys Leu
 35 40 45
 Met Ile Tyr Glu Gly Thr Lys Arg Pro Ser Gly Val Ser Asn Arg Phe
 50 55 60
 Ser Ala Ser Lys Ser Ala Thr Ala Ala Ser Leu Thr Ile Ser Gly Leu
 65 70 75 80
 Gln Pro Glu Asp Glu Ala Asp Tyr Tyr Cys Cys Ser Tyr Ala Thr Ser
 85 90 95
 Arg Thr Leu Val Phe Gly Gly Gly Thr Lys Leu Thr Val Val Gly Gln
 100 105 110
 Pro

<210> SEQ ID NO 68
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 68

Asn Ile Val Met Thr Gln Ser Pro Lys Ser Met Ser Met Ser Val Gly
 1 5 10 15
 Glu Arg Val Thr Leu Thr Cys Lys Ala Ser Glu Asn Val Val Thr Tyr
 20 25 30
 Val Ser Trp Tyr Gln Gln Lys Pro Glu Gln Ser Pro Lys Leu Leu Ile
 35 40 45
 Tyr Gly Ala Ser Asn Arg Tyr Thr Gly Val Pro Asp Arg Phe Thr Gly
 50 55 60
 Ser Gly Ser Ala Thr Asp Phe Thr Leu Thr Ile Ser Ser Val Gln Ala
 65 70 75 80
 Glu Asp Leu Ala Asp Tyr His Cys Gly Gln Gly Tyr Ser Tyr Pro Tyr
 85 90 95

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Thr Phe Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 69
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 69

Gln Ile Val Leu Ser Gln Ser Pro Ala Ile Leu Ser Ala Ser Pro Gly
1 5 10 15

Glu Lys Val Thr Met Thr Cys Arg Ala Ser Ser Ser Val Ser Tyr Ile
20 25 30

His Trp Phe Gln Gln Lys Pro Gly Ser Ser Pro Lys Pro Trp Ile Tyr
35 40 45

Ala Thr Ser Asn Leu Ala Ser Gly Val Pro Val Arg Phe Ser Gly Ser
50 55 60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Arg Val Glu Ala Glu
65 70 75 80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Thr Ser Asn Pro Pro Thr
85 90 95

Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 70
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 70

Asp Ile Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Thr Pro Gly
1 5 10 15

Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser Lys Ser Leu Leu His Ser
20 25 30

Asn Gly Ile Thr Tyr Leu Tyr Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35 40 45

Pro Gln Leu Leu Ile Tyr Gln Met Ser Asn Leu Val Ser Gly Val Pro
50 55 60

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65 70 75 80

Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Ala Gln Asn
85 90 95

Leu Glu Leu Pro Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 71
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Polypeptide

<400> SEQUENCE: 71

Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala Ala Pro Gly Gln
1 5 10 15

Lys Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Asn Asp

20										25					30				
Tyr	Val	Ser	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Thr	Ala	Pro	Lys	Leu	Leu				
		35					40					45							
Ile	Tyr	Asp	Asn	Asn	Lys	Arg	Pro	Ser	Gly	Ile	Pro	Asp	Arg	Phe	Ser				
		50					55					60							
Gly	Ser	Lys	Ser	Gly	Thr	Ser	Ala	Thr	Leu	Gly	Ile	Thr	Gly	Leu	Gln				
65					70					75					80				
Thr	Gly	Asp	Glu	Ala	Asn	Tyr	Tyr	Cys	Ala	Thr	Trp	Asp	Arg	Arg	Pro				
				85					90					95					
Thr	Ala	Tyr	Val	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu					
		100							105					110					

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<210> SEQ ID NO 72
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide
```

<400> SEQUENCE: 72

Gln	Ser	Ala	Leu	Thr	Gln	Pro	Ala	Ser	Val	Ser	Gly	Ser	Pro	Gly	Gln
1				5					10					15	
Ser	Ile	Thr	Ile	Ser	Cys	Thr	Gly	Thr	Ser	Ser	Asp	Val	Gly	Gly	Tyr
			20					25					30		
Asn	Tyr	Val	Ser	Trp	Tyr	Gln	Gln	His	Pro	Gly	Lys	Ala	Pro	Lys	Leu
		35					40					45			
Met	Ile	Tyr	Asp	Val	Ser	Asn	Arg	Pro	Ser	Gly	Val	Ser	Asn	Arg	Phe
	50					55					60				
Ser	Gly	Ser	Lys	Ser	Gly	Asn	Thr	Ala	Ser	Leu	Thr	Ile	Ser	Gly	Leu
65					70					75					80
Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys	Ser	Ser	Tyr	Thr	Ser	Ser
				85					90					95	
Ser	Thr	Arg	Val	Phe	Gly	Thr	Gly	Thr	Lys	Val	Thr	Val	Leu		
			100					105					110		

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<210> SEQ ID NO 73
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Polypeptide
```

<400> SEQUENCE: 73

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Thr	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala	Ser	Ser	Arg	Val	Gly	Tyr	Met
			20					25					30		
His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr
		35					40					45			
Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser
	50					55					60				
Gly	Ser	Gly	Thr	Glu	Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Asp
65					70					75					80
Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Phe	Gln	Gly	Ser	Gly	Tyr	Pro	Phe	Thr
			85						90					95	
Phe	Gly	Gly	Gly	Thr	Lys	Val	Glu	Ile	Lys						
			100					105							

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What is claimed is:

1. An antibody or antigen-binding portion thereof, wherein said antibody or antigen-binding portion thereof binds to an E-DII epitope of Yellow Fever Virus, and wherein the antibody or antigen-binding portion thereof comprises: a) the three CDRs of the heavy chain variable region (VH) as set forth in SEQ ID NO: 4; and the three CDRs of the light chain variable region (VL) as set forth in SEQ ID NO: 38.

2. The antibody or antigen-binding portion thereof of claim 1, wherein the antibody or antigen-binding portion thereof comprises a VH as set forth in SEQ ID NO: 4, and a VL as set forth in SEQ ID NO: 38.

3. The antibody or antigen-binding portion thereof of claim 1, wherein the antibody or antigen-binding portion thereof is a monoclonal antibody, a chimeric antibody, or a humanized antibody.

4. The antibody or antigen-binding portion thereof of claim 1, wherein the antibody is a single chain antibody, a F(ab')₂ fragment, a dAb fragment, a Fab fragment, or a Fv fragment.

5. The antibody or antigen-binding portion thereof of claim 1, wherein the antibody is an isolated antibody.

6. A composition comprising an antibody or antigen-binding portion thereof of claim 1 and a carrier.

7. The composition of claim 6, wherein the composition is a pharmaceutical composition that comprises a pharmaceutically acceptable carrier.

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8. An antibody or antigen-binding portion thereof of claim 1 for use in a medicament.

9. A method for treating Yellow Fever, or a disease or condition associated with Yellow Fever Virus, comprising administering to a subject in need thereof a therapeutically effective amount of an antibody or antigen-binding portion thereof of claim 1.

10. A nucleic acid comprising a nucleotide sequence(s) encoding the antibody or antigen-binding portion thereof of claim 1.

11. A vector comprising the nucleic acid of claim 10.

12. The vector of claim 11, wherein the vector is an expression vector.

13. A host cell comprising the nucleic acid of claim 10.

14. A method of producing an antibody or antigen-binding portion thereof that binds to Yellow Fever Virus E-protein, comprising: culturing the host cell of claim 13 under conditions allowing for expression of the antibody or antigen-binding portion thereof.

15. The method of claim 14, further comprising harvesting the antibody or antigen-binding portion thereof.

16. The method of claim 14, wherein the antibody or antigen-binding portion thereof binds to the E-DII epitope of Yellow Fever Virus E-protein.

17. A host cell comprising the vector of claim 11.

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